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Research Papers

A new two-stage controller design for frequency regulation of low-inertia power system with virtual synchronous generator

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ABSTRACT

This paper proposes a novel two-stage frequency-regulating control design for modern power systems considering high renewable energy sources (RESs) penetration and electric vehicles (EVs). The proposed controller structure is based on a tilt fractional-order integral derivative (TFOID) in the first stage and a proportional derivative with filter (PDN) in the second stage, referred to as a TFOID-PDN controller. Moreover, this paper proposes a virtual synchronous generator (VSG) based on EVs' batteries that emulate the characteristics of synchronous generators and provide inertia and damping properties, thus restraining the frequency instability problem of future power systems when operating in low inertia. Furthermore, the parameters of both the proposed VSG and the TFOID-PDN controller are fine-tuned using a new reliable metaheuristic optimization algorithm called the artificial hummingbird algorithm (AHA). The efficacy of the proposed TFOID-PDN controller design is examined and investigated through a real large multi-source power system (e.g., Egyptian power system) considering the future scenario in 2035 (i.e., in the presence of high penetration levels of RESs, EVs, and different operating conditions). The superiority of the proposed AHA is validated by comparing it with other powerful optimization techniques such as the marine predators algorithm, grey wolf optimizer, and artificial bee colony optimization through designing the load frequency control based on the PID controller of a well-known two-area interconnected power system. Additionally, the proficiency of the proposed controller is verified over other controllers used in the literature, e.g., fractional-order proportional integral derivative (FOPID), tilt integral derivative (TID), proportional integral derivative (PID), and proportional-integral (PI) controllers, under load/RESs fluctuations. The simulation results carried out by the MATLAB software proved the superior performance of the proposed TFOID-PDN controller compared to other controllers. Moreover, the proposed VSG design prevents modern power systems from reaching instability when operating with a high share of RESs and low inertia.

1. Introduction

There is a worldwide interest in reducing the electricity generation from fossil fuels due to the accompanying issues and moving to generation from renewable energy sources (RESs) due to the environmental, economic, and energy independence advantages [1]. However, the high

Abbreviations: ABC, Artificial bee colony; ACE, Area control error; AHA, Artificial hummingbird algorithm; CSP, Concentrated solar power; EPS, Egyptian power system; ESS, Energy storage systems; EV, Electric vehicle; FOPID, Fractional-order proportional integral derivative; GDB, Governor deadband; GRC, Generation rate constraint; GWO, Grey wolf optimizer; ISE, Integral of squared error; ISES, Integrated sustainable energy strategy; LFC, Load frequency control; MGs, Microgrids; MOS, Maximum overshoot; MPA, Marine predators algorithm; MUS, Maximum undershoot; P.F., Participation factor; PI, Proportional integral; PID, Proportional integral derivative; PV, Photovoltaic; RESs, Renewable energy sources; RoCoF, Rate of change of frequency; SG, Synchronous generator; SLP, Step load perturbation; SMES, Superconducting magnetic energy storage; TF, Transfer function; TFOID-PDN, Tilt fractional order integral derivative-proportional derivative with filter; TID, Tilt integral derivative; T_S, Settling time (s); VI, Virtual inertia; VSG, Virtual synchronous generator; WG, Wind generation.

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share of RESs could result in many technical challenges in the operation and control of future power systems because their power generation is intermittent and difficult to predict. Therefore, the technical issues that could be caused by the high share of RESs received great attention from the research community [2].

One of the crucial challenges that have received considerable interest from researchers is the frequency control of power systems with a high share of RESs. Many articles studied frequency control in different power systems (i.e., large power systems and microgrids (MGs)) in the presence of different RESs. For instance, the authors in [3] studied the load frequency control (LFC) of a two-area interconnected power system in the presence of photovoltaic (PV) and wind generation (WG). Moreover, the authors in [4] studied the frequency control issue of a three-area interconnected power system in the presence of WG and solar thermal power plant (STPP). The frequency stability analysis of an islanded MG containing PV and WG has been conducted in [5]. To overcome the high deviations in system frequency that result from RESs fluctuations, many studies investigated the use of different energy storage systems (ESSs) to support frequency control, such as flywheels [6], batteries [7], pumped hydro [8], supercapacitors [9], superconducting magnetic energy storage (SMES) [10], among others. The results of different studies proved the promising role of ESSs could play in the frequency control of different power systems integrated with a high share of RESs. However, their high cost hinders the large-scale adoption of large power systems. Therefore, many studies proposed the use of available controllable devices, such as electric vehicles (EVs), heat pumps, etc., to participate in the frequency control of power systems. The contribution of EVs has been investigated in enhancing the frequency stability of different power systems, such as in an MG [11], two-area interconnected power systems [12], three-area interconnected power systems [8], and others. The results of these studies proved the EVs' effectiveness in providing this ancillary service (i.e., frequency regulation) [13].

Several control techniques/strategies are proposed in the literature for the LFC of power systems, such as proportional integral derivative (PID) [14], fractional-order PID (FOPID) [15], model predictive control [16], fuzzy PID [17], coefficient diagram method [18] among others. These controllers usually have parameters that need to be fine-tuned in the optimum way to achieve the best performance. The selection of optimal parameters for different frequency controllers in power systems is a tedious task. Therefore, many researchers used different optimization techniques to find the optimal parameters of frequency controllers, such as particle swarm optimization (PSO) [19], genetic algorithm [20], bees algorithm [21], Jellyfish Search Optimizer [22], gorilla troops optimization algorithm [23], and many other recently developed optimization algorithms that showed their effectiveness in the LFC applications.

Besides the intermittent nature of RESs that causes many control issues in power systems, they are integrated into the power systems through power electronic converters (i.e., static devices), which have no inertia characteristics, thus reducing the overall inertia of the power system. The reduction in power system inertia results in a large rate of change of frequency (RoCoF) and frequency deviations and could lead to power system instability. The inertia of the physically rotating mass (i.e., generators and motors) is responsible for reducing the RoCoF. One of the proposed approaches to overcome this issue is using virtual synchronous generators (VSG) [24]. In VSG, the power electronic devices of ESSs or RESs are controlled to emulate the characteristics of synchronous generators (SGs) and provide inertia and damping properties that could enhance frequency stability and reduce the RoCoF and frequency deviations after disturbances. This concept is also known in the literature as follows; virtual inertia control, virtual synchronous machine, and synchronverter. Many research works studied the effectiveness of the VSG in enhancing the frequency stability of modern power systems. More interest in these studies was given to MGs [25]. However, other studies considered interconnected power systems [26,27].

Many previous articles studied frequency regulation of real power systems considering different RESs and new flexible devices that are expected to be integrated into the grid, such as Scotland's power system [28], Great Britain's (GB) power system [29], Spain's power system [30], Flinders island in Australia [31], Bornholm island in Denmark [32], and La Palma island in Spain [33]. For the last decades, the Egyptian power system (EPS) has depended on fossil fuels for electricity generation with low shares of RESs, mainly hydropower plants and wind generation. Therefore, the RESs' effect on the operation and control of EPS was negligible. Egypt has ambitious targets to increase the share of RESs in its electricity grid. According to Egypt's integrated sustainable energy strategy to 2035 (ISES 2035) [34], Egypt plans to have 42 % RESs of its generation capacity in 2035. However, these significant EPS changes from deterministic dispatchable fossil fuel-based generation to intermittent non-dispatchable RESs could result in many technical challenges. Therefore, there is a need for many studies considering the 2035 future scenario of the EPS to understand any technical challenges that could be faced and look for solutions to overcome these challenges. Motivated by studying the frequency stability of a real power system integrated with RESs, this study focuses on the frequency stability of the EPS in the 2035 future scenario considering the presence of various RESs and traditional power plants. Additionally, considering the interest of the Egyptian government to upgrade the electric grid towards a smart grid as well as increase the adoption of EVs, the participation of EVs in frequency regulation is studied. Therefore, the EVs are controlled to emulate the characteristics of SGs and operate as VSG to compensate for the low inertia of the power system and enhance frequency stability.

Few studies have addressed the LFC issue for large modern multi-source power systems (e.g., the EPS) considering the high penetration of RESs. For instance, the authors of [35] proposed a digital LFC based on the optimal PID controller to enhance the frequency stability of a real multi-source power system in Egypt. Moreover, the frequency stability analysis of the EPS in the presence of PV and WG has been conducted in [19]. On the other side, several ESSs have been proposed for supporting the frequency stability of modern multi-source power systems (e.g., the EPS). Where the SMES technology has been proposed as a fast response storage system to improve the frequency stability of the EPS considering the high penetration level of WG [36]. However, all these studies didn't consider the 2035 future scenario, and they considered the 2020 scenario, where RESs still do not represent a high share of the generation capacity. Therefore, the authors in [37] have addressed the frequency stability issue for the EPS with the future scenario in 2035, and it was the first study to assess the frequency stability of future EPS. However, it only focused on studying the frequency stability in this future scenario under different operation conditions and comparing the performance of well know controllers in LFC, such as FOPID, PID, PI, and I controllers. Additionally, it did not consider VSG participation in enhancing frequency stability such systems. Therefore, this paper proposes new approaches to improve the frequency stability of EPS in 2035 scenario.

The contributions of this paper are summarized as follows:

1. Propose a new two-stage controller design for the LFC of modern power systems considering high RESs penetration and EVs. The proposed controller structure is based on a tilt fractional-order integral derivative (TFOID) in the first stage and a proportional derivative with filter (PDN) in the second stage, referred to as a TFOID-PDN controller. Moreover, the superior performance of the proposed controller is confirmed by comparing it with other controllers used in the literature (e.g., FOPID, TID, PID, and PI controllers) under load/RESs fluctuations.
2. The proposed TFOID-PDN controller as well as other controllers used in the literature (i.e., FOPID, TID, PID, and PI controllers) are optimally designed using a new optimization algorithm called artificial hummingbird algorithm (AHA) to test its performance in LFC application. Moreover, the proficiency of the proposed AHA is verified over other powerful optimization techniques used in the

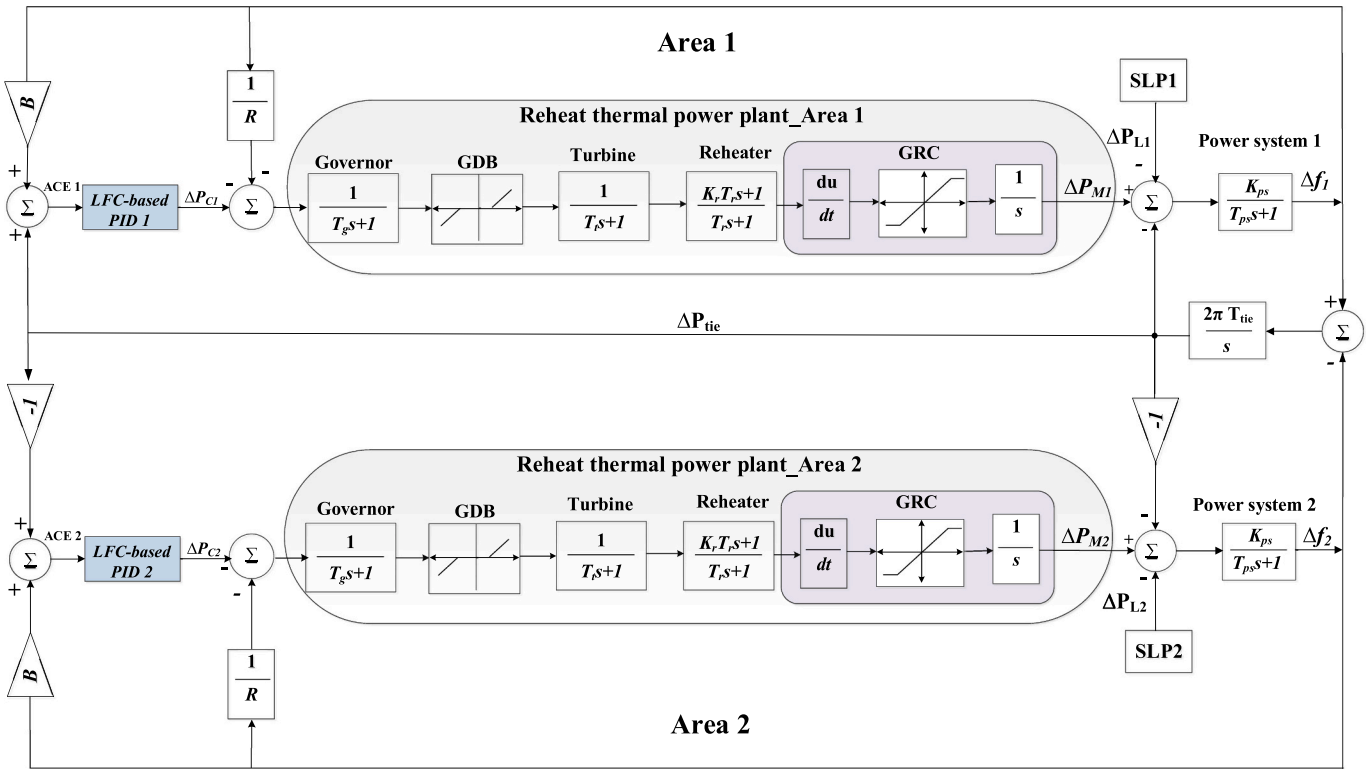


Fig. 1. A dynamic model of the tested two-area interconnected power system.

literature; marine predators algorithm (MPA), grey wolf optimizer (GWO), and artificial bee colony optimization (ABC), through designing the LFC based on PID controller of a well-known two-area interconnected power system.

3. The effectiveness of the proposed controller design is examined and investigated through a real large multi-source power system in Egypt (i.e., the EPS) considering the future scenario in 2035 (i.e., in the presence of high penetration levels of RESs, EVs, and different operating conditions).
4. Enhance the frequency stability of future renewable power systems (e.g., the EPS with the future scenario in 2035) using VSG based on EVs' batteries that emulate the characteristics of SGs and provide inertia and damping properties, thus restraining the frequency instability problem of future power systems when operating in low inertia. Moreover, the parameters of both the proposed VSG and the LFC controller are fine-tuned using the AHA to obtain the minimum system frequency deviation.

The organization of this paper is as follows. Section 2 presents the two-area power system, the proposed optimization algorithm, and a comparison of its performance with other recent optimization algorithms. Section 3 presents the dynamic model of EPS, EVs, and VSG. Section 4 presents the proposed controller and the other controllers. The results and discussion are presented in Section 5, and the conclusion is presented in Section 6.

2. Two-area interconnected reheat power system

In this study, a new metaheuristic optimization algorithm (i.e., the AHA) is proposed for selecting the optimal parameters of the LFC. One of the main weaknesses of metaheuristic optimization algorithms is that they do not perform effectively in all applications. Therefore, the effectiveness of the AHA in the LFC application is compared with other powerful optimization techniques used in the literature; MPA, GWO, and ABC, which are recent and widely used optimization algorithms that have shown good performance in many applications and the LFC specifically.

The two-area interconnected reheat power system is selected as a test system to perform the comparison between optimization algorithms. Additionally, the system's non-linearities, such as governor dead band (GDB) and generation rate constraint (GRC) are considered. In this study, the GDB is 0.06 %, and the GRC is 5 %/min. The two areas of the studied power system are identical; each area contains a reheat thermal power plant. The rated load demand is 1000 MW, the rated power capacity for each area is 2000 MVA, and the power base of the system is 2000 MVA. The dynamic model of the studied two-area interconnected power system is given in Fig. 1, including the transfer functions (TFs) of different components, which are given in (1) to (4). The studied power system parameters are provided in Appendix 1 [38,39].

$$G_{ps}(s) = \frac{K_{ps}}{T_{ps}s + 1} \tag{1}$$

$$G_g(s) = \frac{1}{T_g s + 1} \tag{2}$$

$$G_t(s) = \frac{1}{T_t s + 1} \tag{3}$$

$$G_r(s) = \frac{K_r T_r s + 1}{T_r s + 1} \tag{4}$$

where $G_{ps}(s)$, k_{ps}, and T_{ps} are power system TF, gain, and time constant, respectively. $G_g(s)$ and T_g are governor TF and time constant, respectively. $G_t(s)$ and T_t are turbine TF and time constant, respectively. $G_r(s)$, k_r, and T_r are reheater TF, gain, and time constant, respectively. R is governor speed regulation, B is the area frequency bias factor, and ΔP_L is the change in load power.

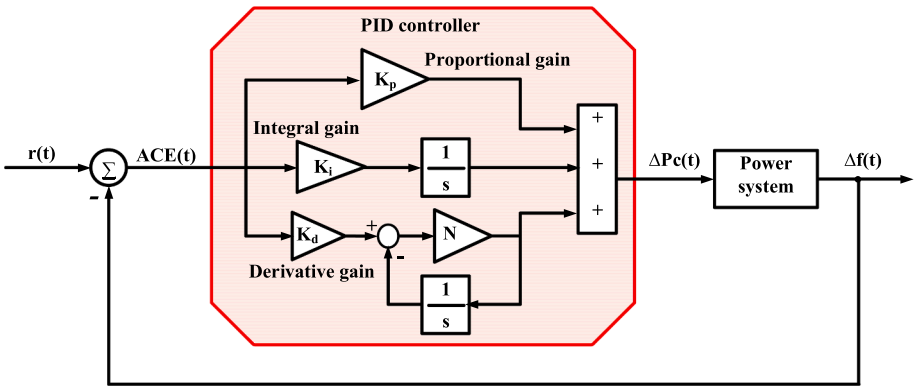


Fig. 2. The PID controller structure.

Table 1
Optimization algorithm settings and variables limits for two area power system.

Parameter	Value
Number of iterations	100
Population size	20
Lower limit of K_p, K_i, and K_d	0.2
Upper limit of K_p, K_i, and K_d	2
Lower limit of N	0
Upper limit of N	100

2.1. LFC-based on PID controller

PID controllers are widely used in the industry due to their good performance and simplicity. Furthermore, they are commonly used in the LFC of power systems. Fig. 2 shows the PID controller structure, and its TF is given in (5). Where $G_{PID}(s)$ is the PID controller TF, K_p is the proportional gain, K_i is the integral gain, K_d is the derivative gain, and N is the derivative filter coefficient. The input to the controller is area control error (ACE) for multi-area power systems or frequency deviation in single-area power systems, and the output is the control signal to generators. To achieve the optimal performance of the PID controller, its four parameters (i.e., K_p, K_i, K_d, and N) must be selected optimally. Many methods are used to fine-tune controller parameters, and recently, many studies showed the effectiveness of metaheuristic optimization algorithms in tuning the parameters of various controllers for LFC applications. In this study, the proposed AHA is tested and validated by finding the optimal parameters of the PID controller.

$$G_{PID}(s) = K_p + \frac{K_i}{s} + K_d \frac{N}{1 + N\frac{1}{s}} \tag{5}$$

In the literature, many performance indices were used as objective functions for tuning controllers of the LFC. In this study, the integral of squared error (ISE) is used as an objective function of the proposed optimization algorithm and is calculated using (6). The main target of the proposed AHA is to find the values of controller parameters that gives the lowest value of ISE, thus obtaining the lowest value of system frequency deviation and tie-line power deviation.

$$ISE = \int_0^{tsim} \left(\Delta f_1^2 + \Delta f_2^2 + \Delta P_{tie}^2\right) dt \tag{6}$$

where t_{sim} is simulation time. Δf_1 and Δf_2 are frequency deviations at area 1 and area 2, respectively. ΔP_{tie} is the power deviation in the tie-line between area 1 and area 2. The PID controllers' parameters should be within limits as given in (7) to (10). The limits of the controller's gains (i.e., K_p, K_i, and K_d) are specified as [0.2, 2], and the limits of N are [0,100]. These limits are specified and extracted from [39] to make a fair comparison between the optimization algorithms used. Since the two areas of the studied power system are identical, the parameters of the two PID controllers are designed to have the same optimal values. Table 1 summarizes the AHA parameters settings and the optimization variables limits that are used to fine-tune the frequency controllers of the studied power systems.

$$K_p^{min} \le K_{p1}, K_{p2} \le K_p^{\max} \tag{7}$$

$$K_i^{min} \le K_{i1}, K_{i2} \le K_i^{\max} \tag{8}$$

$$K_d^{min} \le K_{d1}, K_{d2} \le K_d^{\max} \tag{9}$$

$$N^{min} \le N_1, N_2 \le N^{max} \tag{10}$$

2.2. Artificial hummingbird algorithm (AHA)

In this study, a new optimization algorithm (i.e., AHA) is proposed to fine-tune the parameters of frequency controllers as well as VSG parameters in a low-inertia smart power system. The AHA, which was released before the end of 2021, is a recent bio-inspired optimization method developed for handling engineering optimization problems. It is based on the cunning ways that hummingbirds behave [40]. The AHA uses three foraging strategies: guided foraging, territorial foraging, and migration foraging, in addition to three flying skills: axial, diagonal, and omnidirectional. Fig. 3 shows an AHA's flowchart. A brief description of the algorithm and key equations are provided in the following subsections, and a more in-depth explanation of AHA may be found in [40].

2.2.1. Initialization

A population of n hummingbirds is randomly distributed across n food sources as follows:

$$x_i = Low + r^*(Up - Low)\ i = 1, \cdots, n \tag{11}$$

where x_i indicates the location of the i_{th} food source, which is the solution to a particular problem, Low and Up denote the lower and upper bounds for a given d-dimensional problem, respectively, and r is a random vector in [0,1]. The visit table of food sources is initialized as follows:

$$VT_{i,j} = \begin{cases} 0 & \text{if } i \neq j \\ \text{null} & i = j \end{cases} \quad i = 1, \cdots, n; j = 1, \cdots, n \tag{12}$$

In the case of $i = j$, $VT_{i,j} =$ null shows that a hummingbird is currently ingesting food from a particular source, while in the case of $i \neq j$, $VT_{i,j} = 0$ denotes that the i_{th} hummingbird in the current iteration has just finished visiting the j_{th} food source.

2.2.2. Guided foraging

Each hummingbird flies to the source of nectar that contains most of the nectar. Three different types of flight are used by hummingbirds:

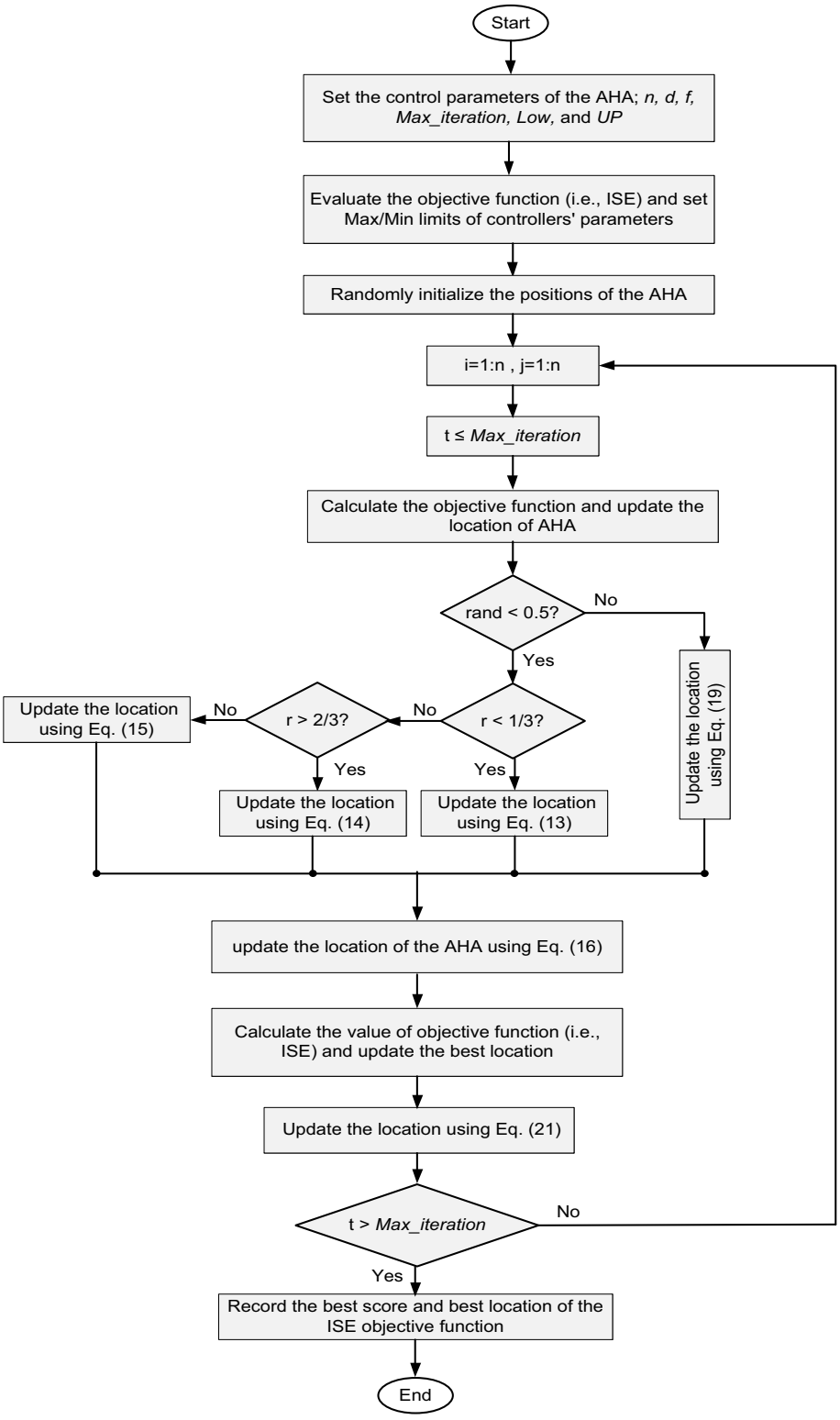


Fig. 3. The flowchart of the AHA.

axial flight, diagonal flight, and omnidirectional flight. Axial flying is described as follows:

$$D^{(i)} = \begin{cases} 1 & \text{if } i = \text{randi}([1,\text{d}]) \\ 0 & \text{else} \end{cases} \; i = 1,\ldots,d \tag{13}$$

The definition of diagonal flight is as follows:

$$D^{(i)} = \begin{cases} 1 & \text{if } i = P(j), j \in [1,k], P = \text{randperm}(k), k \in [2, \lceil r_1 \cdot (d-2) \rceil + 1] \\ 0 & \text{else } i = 1,\cdots,d \end{cases} \tag{14}$$

The definition of omnidirectional flight is as follows:

$$D^{(i)} = 1 \; i = 1,\ldots,d \tag{15}$$

Table 2
Optimal parameters of the PID controller using different optimization algorithms.

Controller	Area 1				Area 2			
	K_p	K_i	K_d	N	K_p	K_i	K_d	N
AHA-PID (proposed)	1.9998	1.9997	1.8793	81.3830	1.9998	1.9997	1.8793	81.3830
MPA-PID [39]	1.79	1.77	1.86	61	1.89	1.83	1.57	59
GWO-PID [39]	1.72	1.71	0.14	87	1.78	1.91	1.83	73
ABC-PID [39]	1.12	1.52	1.72	71	1.92	1.37	0.68	59

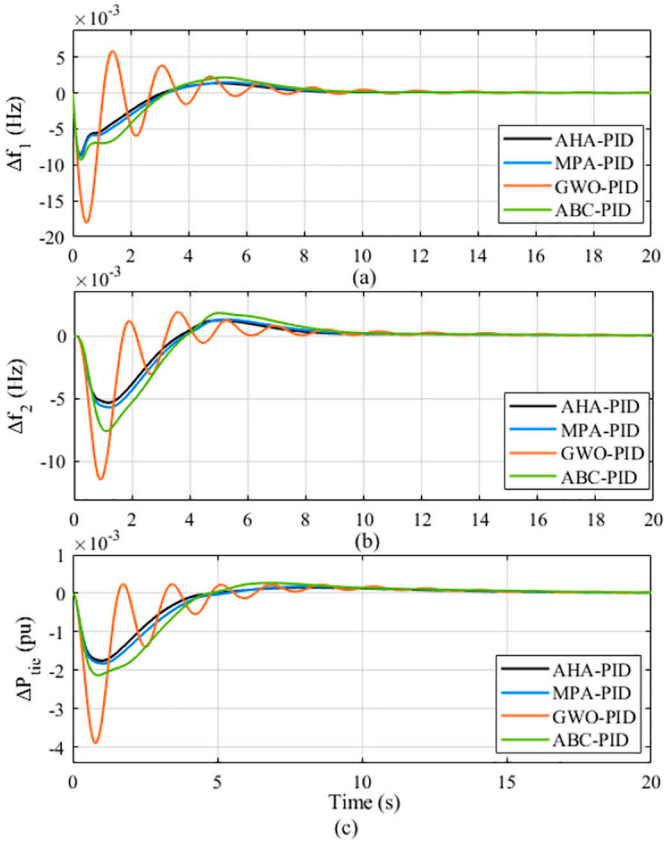


Fig. 4. System response for case 1. (a) Δf_1, (b) Δf_2, and (c) ΔP_{tie}.

where; *randi*([1, *d*]) creates a random number between 1 and d, randperm (*k*) creates a permutation of numbers from 1 to k, and r_1 is a random number [0, 1]. The mathematical formula for simulating directed foraging behavior with a suitable food source is as follows:

$$v_i(t+1) = x_{i,tar}(t) + a \cdot D \cdot \left(x_i(t) - x_{i,tar}(t)\right) \tag{16}$$

$$a \sim N(0,1) \tag{17}$$

where; the i_{th} food source's location at time t is represented by $x_i(t)$, $x_{i,tar}(t)$ is the location of the food supply which i_{th} hummingbird intends to visit it, and a is an indicative factor that is subject to a normal distribution $N(0,1)$ with a standard deviation of 1 and a mean of 0.

The following is an update on the location of the i_{th} food source:

$$x_i(t+1) = \begin{cases} x_i(t) & f(x_i(t)) \le f(v_i(t+1)) \\ v_i(t+1) & f(x_i(t)) \rangle f(v_i(t+1)) \end{cases} \tag{18}$$

According to Eq. (18), the hummingbird abandons the current food source and feeds at the candidate food source identified by Eq. (16) if the candidate food source's nectar-refilling rate is higher than the current food source.

2.2.3. Territorial foraging

A hummingbird can readily relocate within its territory to a neighboring location where it can find a new food source that may be superior to the one it now has. The local foraging strategy for hummingbirds and an appropriate food source in their territorial foraging strategy are described by the mathematical equation below:

$$v_i(t+1) = x_i(t) + b \cdot D \cdot x_i(t) \tag{19}$$

$$b \sim N(0,1) \tag{20}$$

where b is a regional factor with a standard deviation of 1 and a mean of 0 as well as is subject to a normal distribution of $N(0,1)$.

2.2.4. Migration foraging

The migration of a hummingbird from a nectar source with the lowest nectar replenishment rate to randomly newly created nectar sources can be explained as follows.

$$x_{wor}(t+1) = Low + r^*(Up - Low) \tag{21}$$

Here, x_{wor} is the food with the lowest nectar replenishment rate among the population.

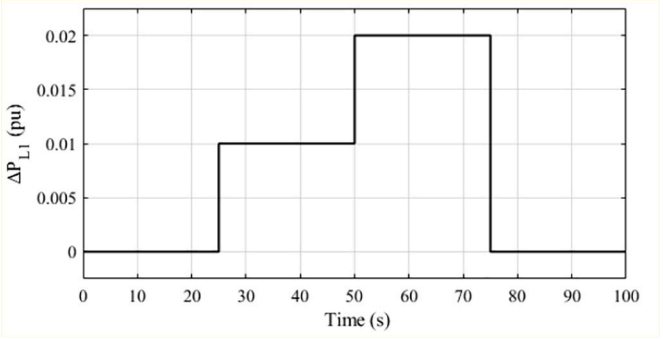


Fig. 5. Random SLP in area 1 of the studied system for case 2.

Table 3
Performance indices values for case 1.

Controllers	Δf_1			Δf_2			ΔP_{tie}		
	MUS (Hz)	MOS (Hz)	T_S (s)	MUS (Hz)	MOS (Hz)	T_S (s)	MUS (pu)	MOS (pu)	T_S (s)
AHA-PID (proposed)	8.600e-3	1.386e-3	15.98	5.346e-3	1.235e-3	17.133	1.750e-3	1.513e-4	14.49
MPA-PID [39]	8.823e-3	1.496e-3	16.64	5.716e-3	1.290e-3	17.97	1.826e-3	1.735e-4	15.35
GWO-PID [39]	1.807e-2	5.837e-3	23.95	1.150e-2	1.902e-3	22.43	3.895e-3	2.439e-4	16.43
ABC-PID [39]	9.299e-3	2.173e-3	16.64	7.621e-3	1.822e-3	18.00	2.130e-3	2.678e-4	16.35

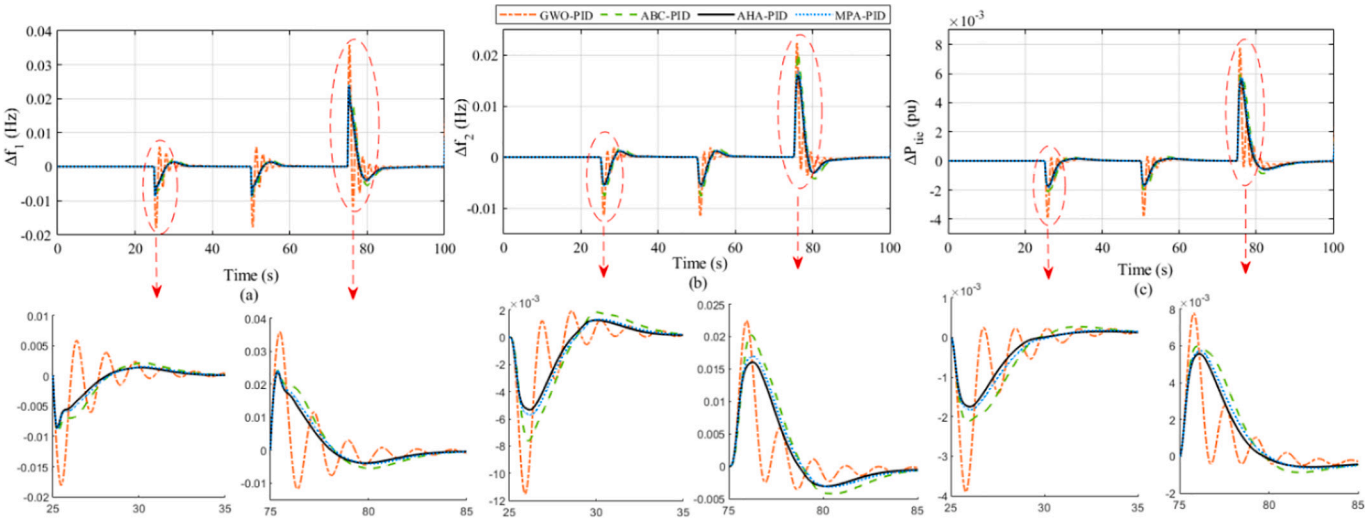


Fig. 6. System response for case 2. (a) Δf_1, (b) Δf_2, and (c) ΔP_{tie}.

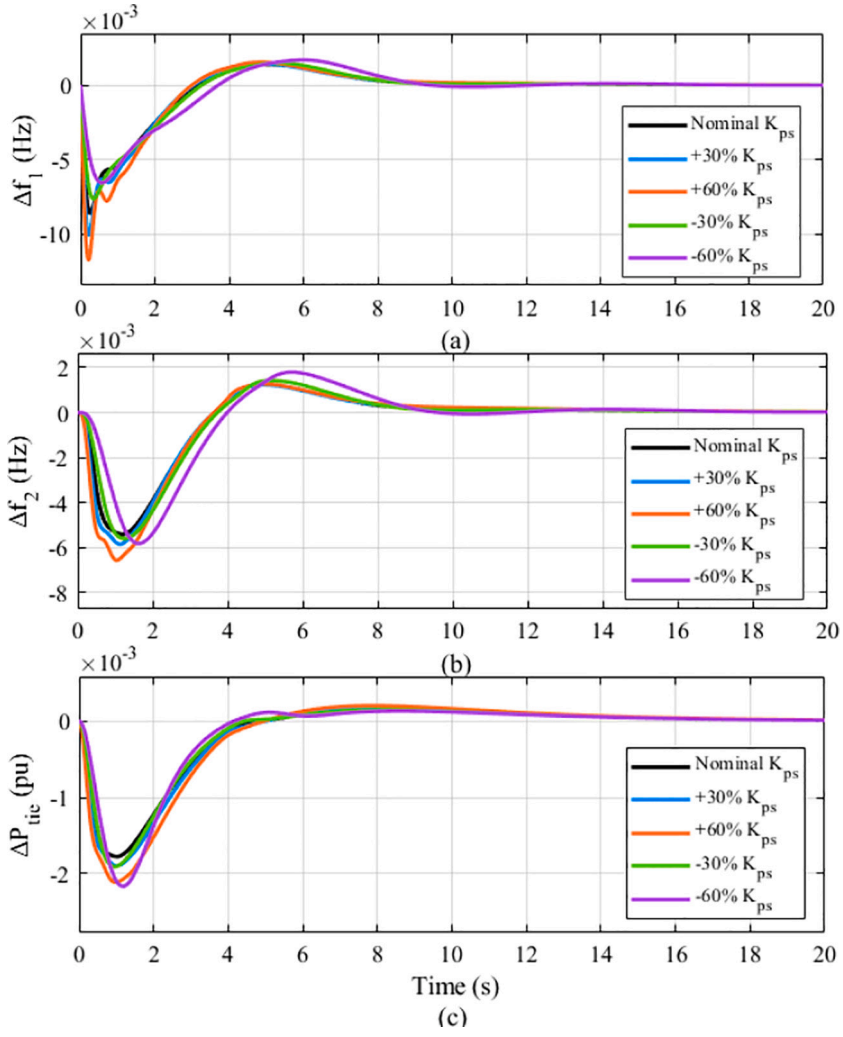


Fig. 7. System response for case 3. (a) Δf_1, (b) Δf_2, and (c) ΔP_{tie}, under changes in K_{ps}.

If there is no replacement for any food sources, a hummingbird practicing directed foraging and territorial foraging would visit each food source in turn as its target source according to the visit table at each iteration. Assume that there is a 50 % chance of success when deciding between guided and territorial foraging and that there is a 50 % chance of success when visiting every other source during guided foraging. In this situation, it is necessary to extend the search region and alleviate immobility by adopting a migratory foraging strategy. Therefore, it is

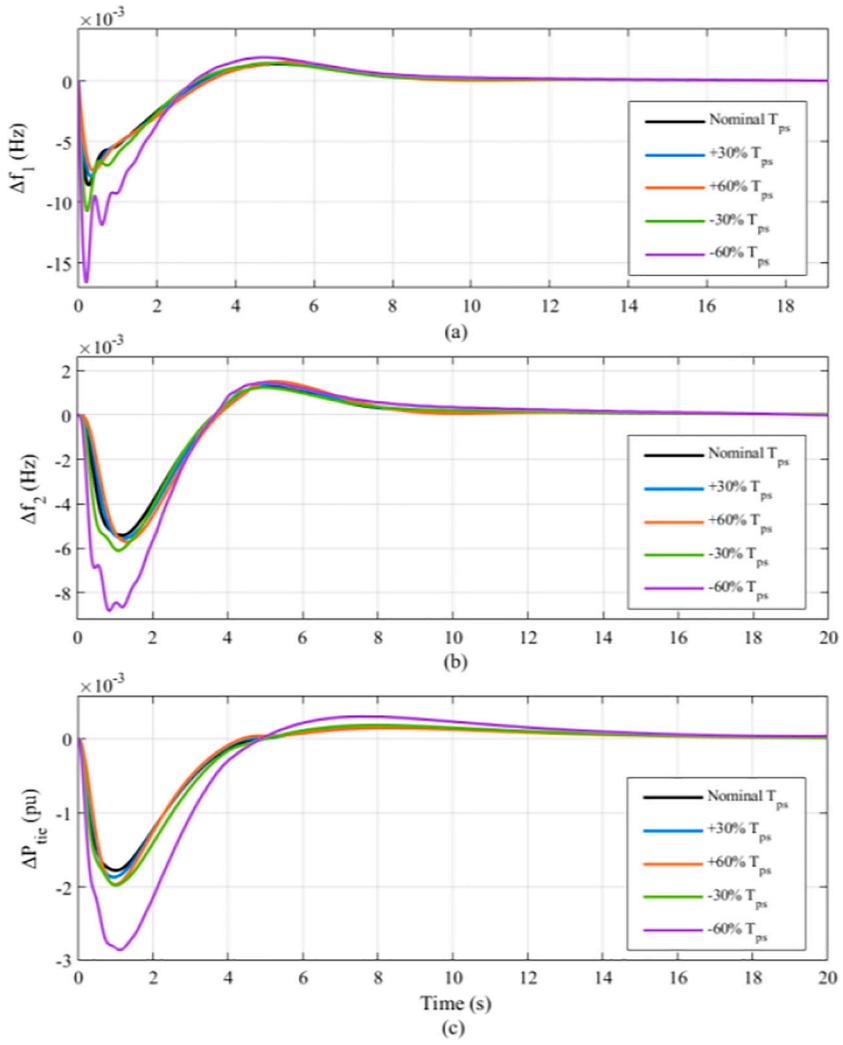


Fig. 8. System response for case 3. (a) Δf_1, (b) Δf_2, and (c) ΔP_{tie}, under changes in T_{ps}.

suggested that the population size specification for the migration coefficient be as follows:

$$M = 2n \tag{22}$$

2.3. Results and discussion

A 1 % step load perturbation (SLP) is applied at area 1, and the proposed AHA optimization algorithm is used to tune the parameters of the PID controllers to minimize the objective function given in eq. (6) in an iterative process until the maximum number of iterations is reached. The optimal parameters are given in Table 2. Additionally, the table shows the optimal parameters found using other recent optimization algorithms (i.e., MPA, GWO, and ABC) given in [39]. In this subsection, the performance of the AHA-based PID controller (i.e., AHA-PID) is compared with the MPA-based PID controller (i.e., MPA-PID) [39], GWO-based PID controller (i.e., GWO-PID) [39], and ABC-based PID controller (i.e., ABC-PID) [39] considering different disturbances and operating conditions.

Case 1. Performance evaluation of the studied system under 1 % step load change in area 1.

In this case, a 1 % SLP is applied in area 1, and the change of Δf_1, Δf_2, and ΔP_{tie} are recorded. The change in their values after the load increase is given in Fig. 4. Additionally, the corresponding settling time (T_S) based on ±0.005 % criterion, maximum undershoot (MUS), and maximum overshoot (MOS) are provided in Table 3. The simulation results show that the proposed AHA-PID controller performance is the best among other controllers tuned using MPA, GWO, and ABC. Moreover, it is clear that the AHA-PID controller achieved lower MUS, MUS, and Ts than the other controllers, as indicated in Table 3.

Case 2. Performance evaluation of the studied system under random load change in area 1.

In this case, a random SLP is applied in area 1 of the studied system. Fig. 5 shows the variation of area 1 load during the simulation period. Fig. 6 shows the variations of Δf_1, Δf_2, and ΔP_{tie} during these disturbances. The results show that the dynamic response of the proposed AHA-PID controller is better than MPA-PID, GWO-PID, and ABC-PID controllers. The AHA-PID controller showed a fast settling time and minor deviations in Δf_1, Δf_2, and ΔP_{tie}.

Case 3. Performance evaluation of the studied system under system parameters variations.

To test the robustness of the proposed AHA-PID controller, a sensitivity analysis is performed where the three parameters (i.e., K_{ps}, T_{ps}, and T_{tie}) of the studied power system are changed by ±30 % and ± 60 % from their nominal values. For all the cases, a 1 % SLP is applied in area 1 of the studied power system. Fig. 7 shows the dynamic response of Δf_1, Δf_2, and ΔP_{tie} for changes in K_{ps}. Fig. 8 shows the dynamic response of Δf_1, Δf_2, and ΔP_{tie} for changes in T_{ps}. Fig. 9 shows the dynamic response of Δf_1, Δf_2, and

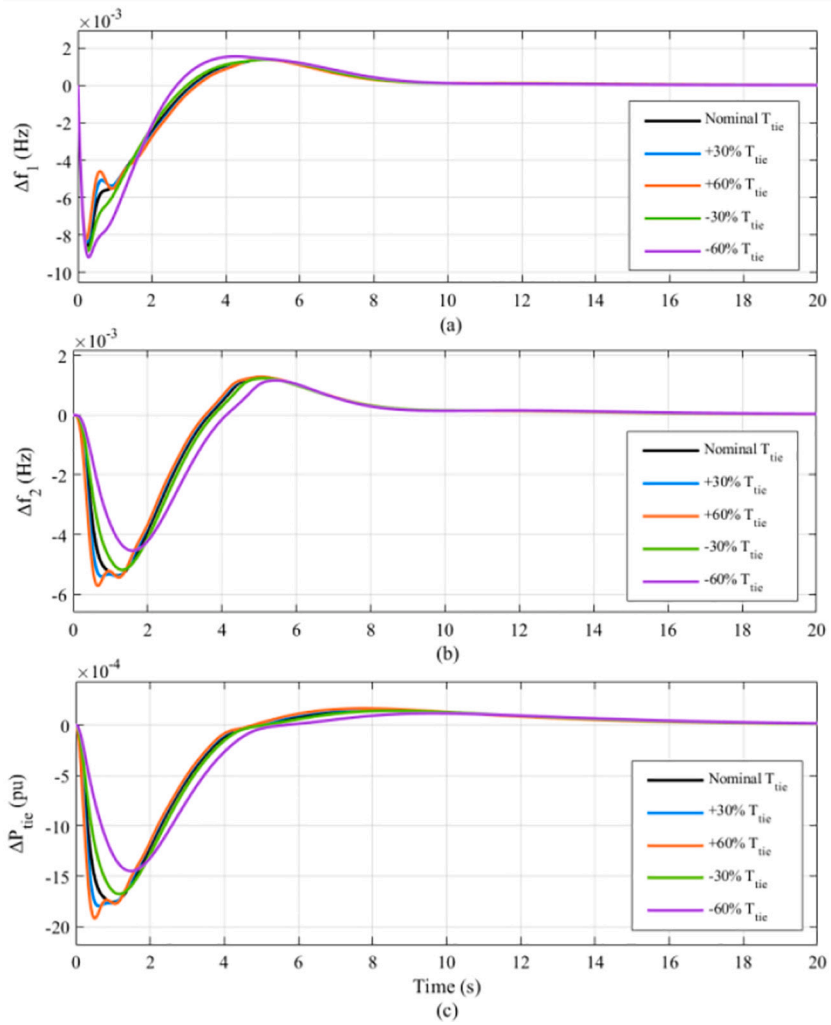


Fig. 9. System response for case 3. (a) Δf_1, (b) Δf_2, and (c) ΔP_{tie}, under changes in T_{tie}.

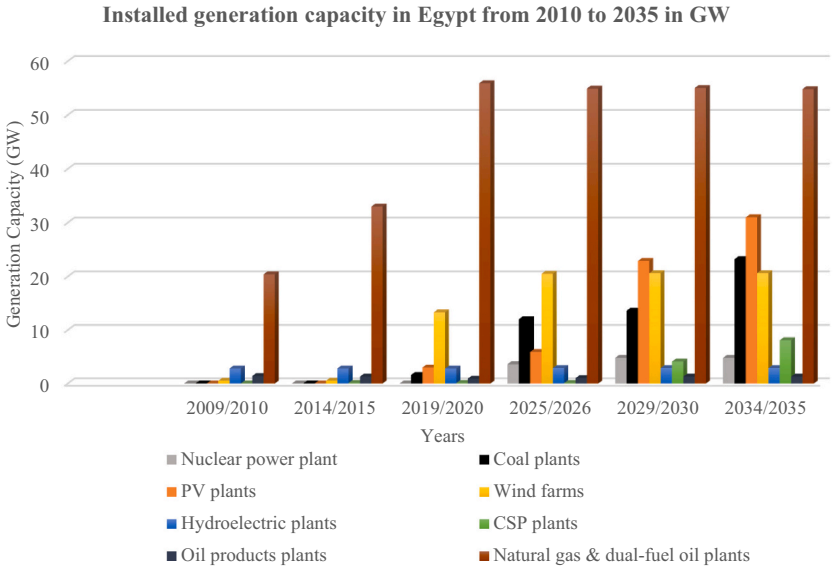


Fig. 10. Growth of Egypt's installed generation capacity, expressed in GW, between 2010 and 2035.

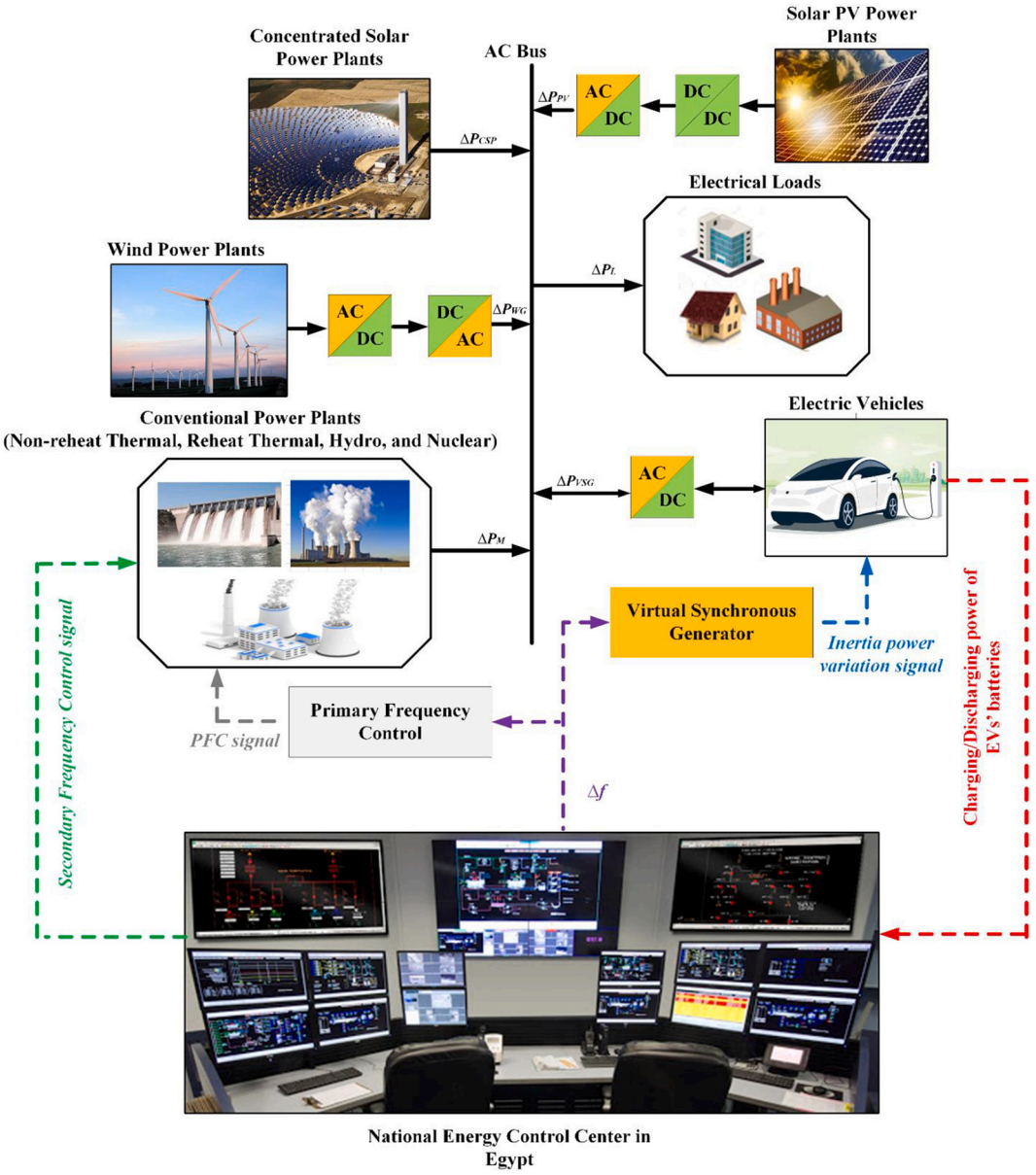


Fig. 11. Schematic diagram of the EPS with the VSG.

ΔP_{tie} for changes in T_{tie}. The results proved the robustness of the proposed AHA-PID controller and that it has a good performance with changes in power system parameters without tuning its parameters again.

From the results of the three studied cases, it could be concluded that the new optimization AHA is effective in the LFC application. Therefore, the proposed AHA algorithm has become trustworthy to be used in designing and finding the optimal parameters for both the proposed TFOID-PDN controller and VSG in modern power systems, e.g., the EPS with the future scenario in 2035.

3. Configuration and modeling of the Egyptian power system

Egypt has ambitious targets in increasing the share of RESs in its electric grid (i.e., the EPS). According to ISES 2035 [34], Egypt plans to have 42 % RESs of its generation capacity in 2035. Fig. 10 shows the targeted growth of the generation capacity of EPS from 2010 to 2035. The total generation capacity in 2035 is targeted to be 146.7 GW. This power will be generated from nuclear power plants and fossil fuels-based power plants that operate using natural gas, coal, or oil. Additionally, a high share of this power will be generated from RESs such as hydropower plants, wind generation (WG) plants, photovoltaic (PV) plants, and concentrated solar power (CSP) plants. In 2035, 2.9 GW will be generated from hydropower plants, 20.6 GW will be generated from WG, 31 GW will be generated from PV, 8.1 GW will be generated from CSP, 39.65 GW will be generated from non-reheat thermal plants, 39.65 GW will be generated from reheat thermal plants, 4.8 GW will be generated from nuclear plants.

Fig. 11 shows a schematic diagram of the EPS, and it shows different types of power plants that will be installed in addition to loads and EVs operating as VSG. The EPS dynamic model is shown in Fig. 12. The EPS is modeled using transfer functions which is considered suitable for studying the power system frequency stability. Furthermore, the system's non-linearities; i.e. GDB and GRC, are considered. For non-reheat power plants, the GDB is 0.06 %, and the GRC is 20 %/min. For reheat power plants, the GDB is 0.06 %, and the GRC is 10 %/min. For nuclear power plants, the GDB is 0.06 %, and the GRC is 5 %/min. For hydropower plants, the GDB is 0.01 %, and the GRC is 270 %/min for increasing generation and 360 %/min for decreasing generation. In this study,

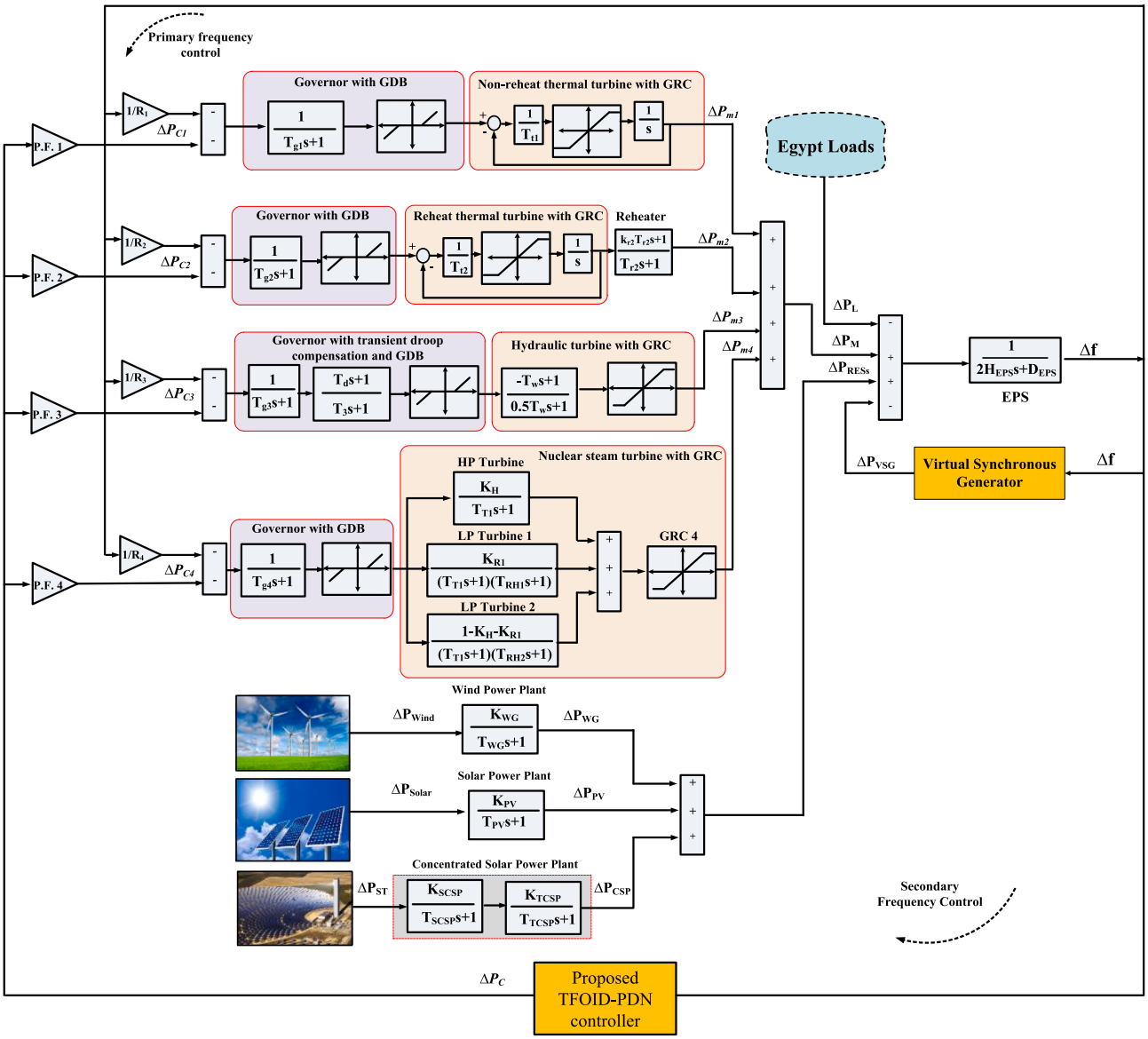


Fig. 12. Block diagram of the Egyptian power system based on the future 2035 scenario.

intermittent RESs do not participate in frequency regulation. Therefore, intermittent RESs (i.e., ΔP_{WG}, ΔP_{PV}, and ΔP_{CSP}) and loads (i.e., ΔP_L) are the sources of disturbances in the studied scenarios, while traditional power plants (i.e., thermal, nuclear, and hydropower plants) and EVs are responsible for frequency control and keeping the frequency within acceptable limits under various disturbances and operating conditions.

3.1. Dynamic model of conventional power plants

For many decades, electricity was generated from traditional power plants operating with fossil fuels, nuclear energy, and water flow in dams. These power plants are reliable, and their power generation can be controlled to follow the variable demand. In the studied scenarios of EPS, traditional power plants are responsible for frequency regulation. In this study, the primary and secondary frequency control loops are considered. The power change of non-reheat thermal power plants ΔP_{m1} [38], reheat thermal power plants ΔP_{m2} [38], hydropower plants ΔP_{m3} [41,42], and nuclear power plants ΔP_{m4} [43] are given in (23)-(26), respectively. ΔP_{C1}, ΔP_{C2}, ΔP_{C3}, and ΔP_{C4} are the control signals sent from the secondary controller to the power plants. The meaning of the symbols in the EPS dynamic model and the values of EPS parameters are provided in Table 4 [37].

$$\Delta P_{m1} = \frac{1}{T_{g1}s+1} * \frac{1}{T_{t1}s+1} * \left(\frac{-1}{R_1} * \Delta f - \Delta P_{c1}\right) \tag{23}$$

$$\Delta P_{m2} = \frac{1}{T_{g2}s+1} * \frac{1}{T_{t2}s+1} * \frac{K_{r2}T_{r2}s+1}{T_{r2}s+1} * \left(\frac{-1}{R_2} * \Delta f - \Delta P_{c2}\right) \tag{24}$$

$$\Delta P_{m3} = \frac{1}{T_{g3}s+1} * \frac{T_d s+1}{T_3 s+1} * \left(\frac{-T_w s+1}{0.5 * T_w s+1}\right) * \left(\frac{-1}{R_3} * \Delta f - \Delta P_{c3}\right) \tag{25}$$

$$\Delta P_{m4} = \frac{1}{T_{g4}s+1} * \left(\frac{K_H}{T_{T1}s+1} + \frac{K_{R1}}{(T_{T1}s+1)(T_{RH1}s+1)} + \frac{1-K_H-K_{R1}}{(T_{T1}s+1)(T_{RH2}s+1)}\right) * \left(\frac{-1}{R_4} * \Delta f - \Delta P_{c4}\right) \tag{26}$$

Table 4
Egyptian power system parameters. Adapted from [37].

Egyptian power system parameters		
H_{EPS}	EPS inertia constant (s)	4.07
D_{EPS}	EPS damping coefficient (pu MW/Hz)	0.028
K_{r2}	Reheater gain of reheat thermal power plants	0.50
K_H	High-pressure turbine gain of nuclear power plants	2.00
K_{R1}	Low-pressure turbine gain of nuclear power plants	0.30
K_{PV}	Photovoltaic generation gain	1.00
K_{SCSP}	Solar collector gain of concentrated solar power plants	1.80
K_{TCSP}	Turbine gain of concentrated solar power plants	1.00
K_{WG}	Wind generation gain	1.00
P.F.1	Participation factor of non-reheat thermal power plants	0.45
P.F.2	Participation factor of reheat thermal power plants	0.45
P.F.3	Participation factor of hydropower power plants	0.04
P.F.4	Participation factor of nuclear power plants	0.06
R_1	Governor speed regulation of non-reheat thermal power plants (Hz/pu MW)	2.50
R_2	Governor speed regulation of reheat thermal power plants (Hz/pu MW)	2.50
R_3	Governor speed regulation of hydropower plants (Hz/pu MW)	1.00
R_4	Governor speed regulation of nuclear power plants (Hz/pu MW)	2.40
T_{g1}	Governor time constant of non-reheat thermal power plants (s)	0.08
T_{t1}	Turbine time constant of non-reheat thermal power plants (s)	0.30
T_{g2}	Governor time constant of reheat thermal power plants (s)	0.08
T_{t2}	Turbine time constant of reheat thermal power plants (s)	0.30
T_{r2}	Reheat time constant of reheat thermal power plants (s)	10.00
T_{g3}	Governor main servo time constant of hydro power plants (s)	0.20
T_3	Governor transient droop time constant of hydropower plants (s)	28.50
T_d	Governor reset time constant of hydropower plants (s)	5.00
T_w	Water starting time in penstock of hydropower plants (s)	1.00
T_{g4}	Governor time constant of nuclear power plants (s)	0.08
T_{T1}	High-pressure turbine time constant of nuclear power plants (s)	0.50
T_{RH1}	First low-pressure turbine time constant of nuclear power plants (s)	7.00
T_{RH2}	Second low-pressure turbine time constant of nuclear power plants (s)	9.00
T_{PV}	Time constant of photovoltaic generation (s)	1.85
T_{SCSP}	Solar collector time constant of concentrated solar power plants (s)	1.80
T_{TCSP}	Turbine time constant of concentrated solar power plants (s)	0.30
T_{WG}	Time constant of wind generation (s)	1.50
T_{EV}	Average time constant of EV chargers (s)	1.00

3.2. Dynamic model of intermittent RESs

Traditional power plants are reliable and dispatchable and therefore result in simple power system operation and control. However, it faces many challenges and issues, such as the depletion of fossil fuels, environmental impacts of fossil fuels, risks of nuclear plants, and limited locations suitable for hydropower plants. Therefore, electricity generation from RESs is indispensable. Electricity generation from RESs received considerable interest, and there is a quick growth of RESs installations worldwide. RESs provide many advantages, but it makes the power system operation and control more challenging because of their intermittent nature and the dependence of power generated on environmental conditions (i.e., wind speed, solar irradiance, etc.). According to ISES 2035, the EPS will contain WG, PV, and CSP.

Egypt has many high-wind-speed locations suitable for electricity generation from the wind, such as the Gulf of Suez, Sinai, New Valley governorate, and a few locations beside the Nile river. Currently, Egypt generates 1375 MW from WG and aims to generate 20.6 GW in 2035. Moreover, electricity generation from PV is very promising in Egypt due to the high solar irradiance during the whole year and the large free land that could be used for PV plant installations. Currently, Egypt generates 1465 MW from PV plants and aims to generate 31 GW in 2035. Furthermore, the sun's energy could be captured in the form of heat using CSP plants. Egypt seeks to generate 8.1 GW in 2035 from CSP plants. The power change of WG (i.e., ΔP_{WG}) [44], PV (i.e., ΔP_{PV}) [44],

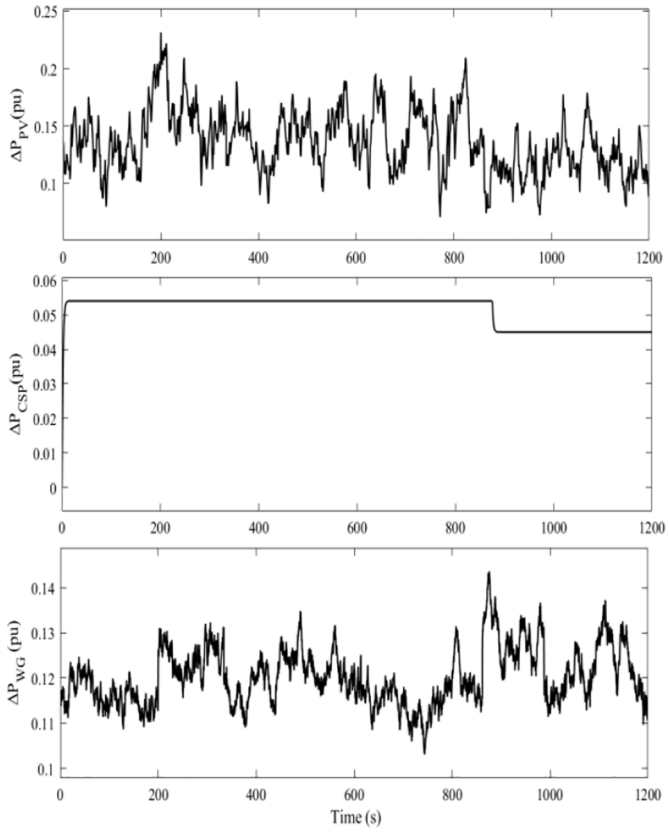


Fig. 13. Power variation of PV, CSP, and WG.

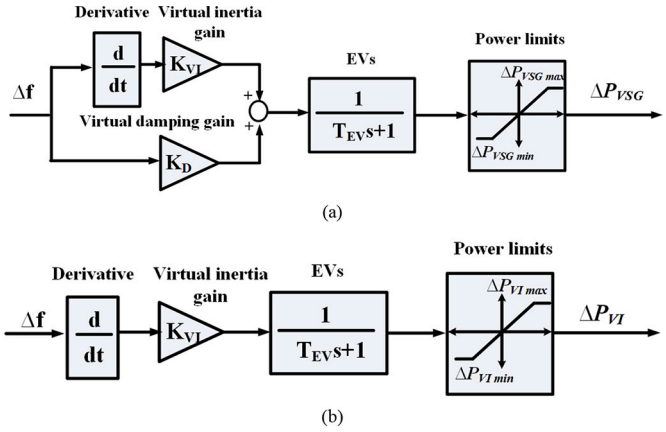


Fig. 14. Virtual synchronous generator based on EVs' batteries; (a) inertia-damping emulation model, and (b) inertia emulation model.

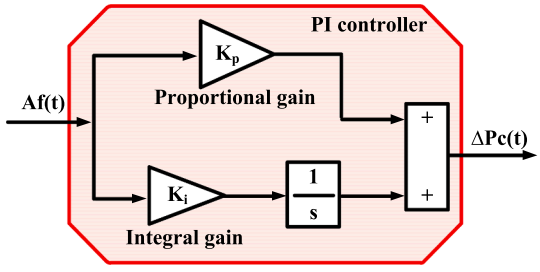


Fig. 15. Structure of the PI controller.

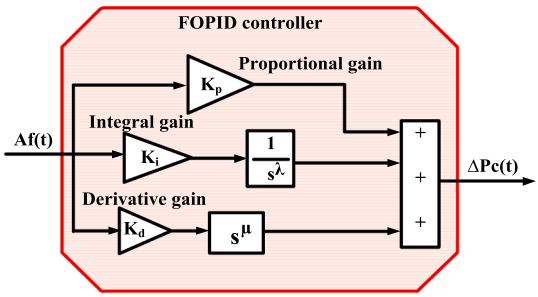


Fig. 16. Structure of the FOPID controller.

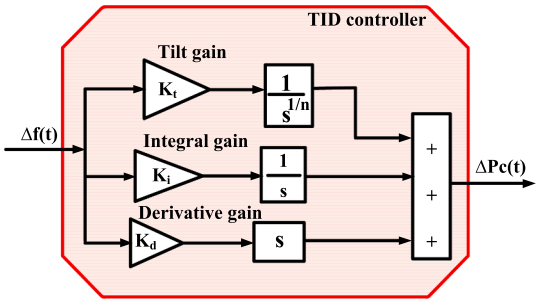


Fig. 17. Structure of the TID controller.

and CSP (i.e., ΔP_{CSP}) [45] are given in (27)-(29), respectively. Fig. 13 shows the power generation of PV, CSP, and WG power plants during the simulation period.

$$\Delta P_{WG} = \left(\frac{K_{WG}}{T_{WG}s+1}\right)*\Delta P_{\text{Wind}} \tag{27}$$

$$\Delta P_{PV} = \left(\frac{K_{PV}}{T_{PV}s+1}\right)*\Delta P_{\text{Solar}}. \tag{28}$$

$$\Delta P_{CSP} = \frac{K_{SCSP}}{T_{SCSP}s+1}*\frac{K_{TCSP}}{T_{TCSP}s+1}*\Delta P_{ST} \tag{29}$$

3.3. Dynamic model of virtual synchronous generator (VSG)

Synchronous generators (SGs) inherit many desired features that kept the electric power systems running reliably for decades. One important feature is the inertia energy that is stored in the rotating mass. The SG's inertia has significant importance in reducing RoCoF after disturbance and enhancing frequency stability. This feature is not present in RESs that are connected to the grid through power electronic converters and lack rotating mass. Therefore, future power systems with a high share of RESs are expected to have low inertia constant, resulting in high-frequency deviation and high RoCoF, which could lead to frequency instability and blackouts. To overcome this issue, the VSG concept is proposed to increase the inertia of low-inertia power systems. This could be achieved by adding a control loop to ESSs power electronics converters to emulate the characteristics of real SGs. Due to the

Table 5
Controllers variables limits for Egyptian power system.

Controller	Parameter	Value
PI	Lower limit of K_p and K_i	0
	Upper limit of K_p and K_i	10
PID	Lower limit of K_p, K_i, K_d, and N	0
	Upper limit of K_p, K_i, K_d, and N	10
FOPID	Lower limit of K_p, K_i, and K_d	0
	Upper limit of K_p, K_i, and K_d	10
	Lower limit of λ and μ	0
	Upper limit of λ and μ	1
TID	Lower limit of K_t, K_i, and K_d	0
	Upper limit of K_t, K_i, and K_d	10
	Lower limit of n	1
	Upper limit of n	10
TFOID-PDN	Lower limit of K_t, K_i, K_{d1}, K_p, K_{d2}, and N	0
	Upper limit of K_t, K_i, K_{d1}, K_p, K_{d2}, and N	10
	Lower limit of n	1
	Upper limit of n	10
	Lower limit of λ and μ	0
	Upper limit of λ and μ	1

Table 6
Optimal parameters of the proposed controllers.

Controller	TFOID-PDN (Proposed)	FOPID	TID	PID	PI
Parameters	$K_t = 1.958$	$K_p =$ 9.9997	$K_t =$ 9.9983	$K_p =$ 9.9968	$K_p =$ 9.9999
	$n = 5.0398$	$K_i =$ 9.9930	$n =$ 6.5001	$K_i =$ 9.9947	$K_i =$ 9.8301
	$K_i = 6.6366$	$K_d =$ 0.9483	$K_i =$ 9.9982	$K_d =$ 9.9953	–
	$\lambda = 0.6755$	$\lambda =$ 9.9821	$K_d =$ 9.9926	$N =$ 1.5609	–
	$K_{d1} = 0.9518$	$\mu =$ 0.3011	–	–	–
	$\mu = 0.9838$	–	–	–	–
	$K_p = 6.2602$	–	–	–	–
	$K_{d2} = 9.0267$	–	–	–	–
	$N = 9.9843$	–	–	–	–
ISE	**2.317e-4**	6.513e-4	9.412e-4	8.698e-4	15.206e-4

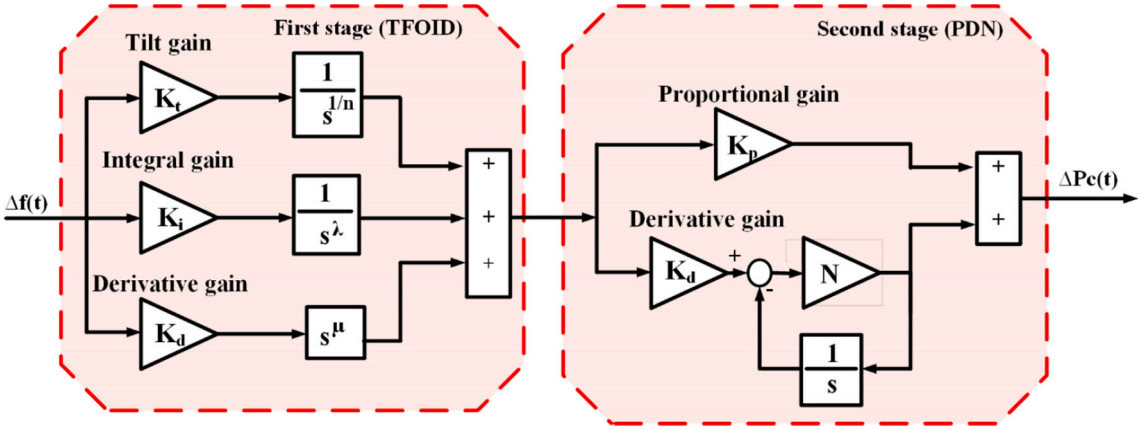


Fig. 18. Structure of the proposed TFOID-PDN controller.

high cost of ESSs, their large-scale adoption could take a long time. On the other hand, EVs adoption is growing very fast, and a large number of EVs are targeted in the near future. The uncontrolled charging of EVs could cause many issues in the power system [46,47], and by the use of smart charging, many ancillary services could be provided [48–50]. Therefore, this study proposes using EVs to emulate SGs and provide frequency stability support by properly controlling their chargers. This study uses a VSG model that is based on the swing equation [51]. The model has two parts, one to emulate inertia and the other to emulate damping properties. The power change of EVs-based VSG is given in (30) and is presented graphically in Fig. 14-a. The model also contains the dynamic model of EV chargers and limits of power capacity [30]. The limits of VSG power capacity are ±0.05 pu. Another common model in the literature is to emulate the inertia properties of SGs. This model does not emulate damping properties. The power change of EVs-based virtual inertia (EV-VI) is given in (31) and is presented graphically in Fig. 14-b.

$$\Delta P_{VSG} = \frac{1}{T_{EV}s + 1} * (K_{VI}s + K_D) * \Delta f \tag{30}$$

$$\Delta P_{VI} = \frac{1}{T_{EV}s + 1} * (K_{VI}s) * \Delta f \tag{31}$$

Based on the described dynamic models of different EPS components (i.e., traditional power plants, intermittent RESs, VSG, and loads) presented in this section, the frequency deviation of the EPS can be calculated using (32).

$$\Delta f = \frac{1}{2Hs + D}(\Delta P_{m1} + \Delta P_{m2} + \Delta P_{m3} + \Delta P_{m4} + \Delta P_{WG} + \Delta P_{PV} + \Delta P_{CSP} + \Delta P_{VSG} - \Delta P_L) \tag{32}$$

4. Proposed controller and problem formulation

To test the performance of the proposed two-stage controller, its performance in the LFC application is compared with PI, PID, FOPID, and TID controllers. This section describes the proposed controller and the other four controllers.

4.1. PID and PI controllers

The PID controller with a filter at the derivative branch is discussed in Section 2.1. The PI controller is a type of PID controller that considers the proportional and integral loops only and therefore has two gains, the proportional gain K_p and the integral gain K_i. The PI controller transfer

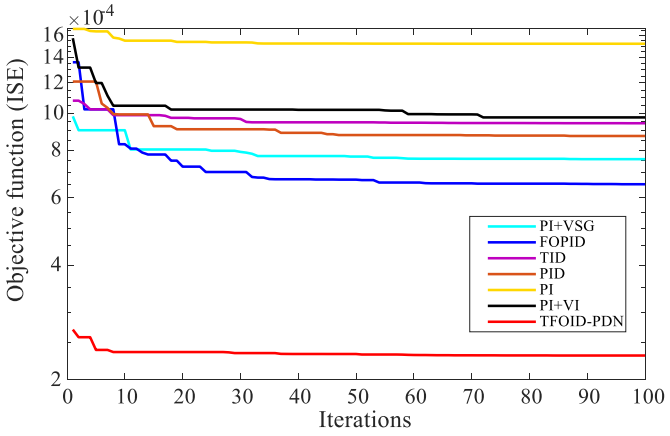


FIG. 19. Convergence curves of AHA for all controllers.

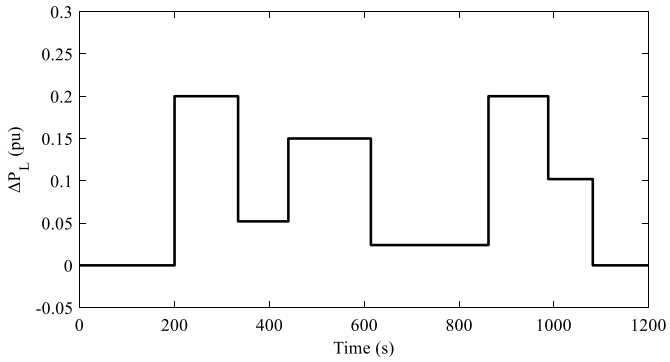


Fig. 21. Random SLP of the load in EPS for scenario 2.

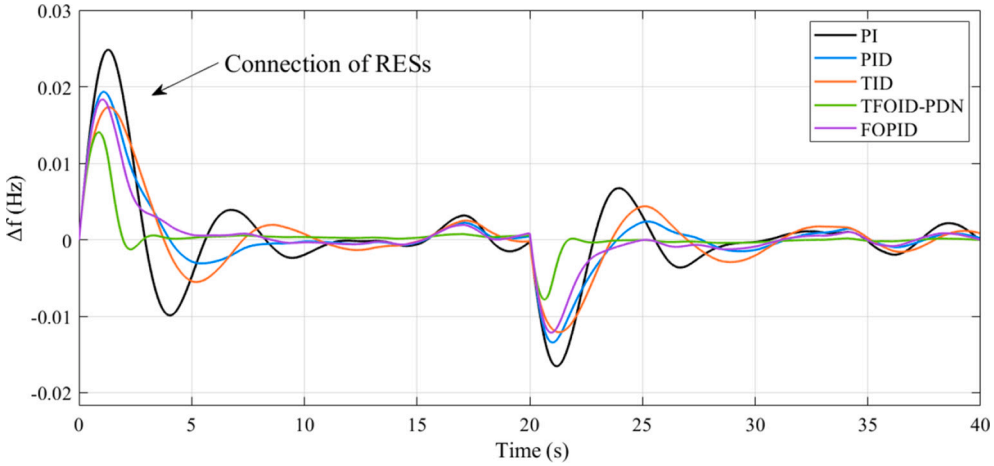


Fig. 20. System frequency deviation of EPS for scenario 1.

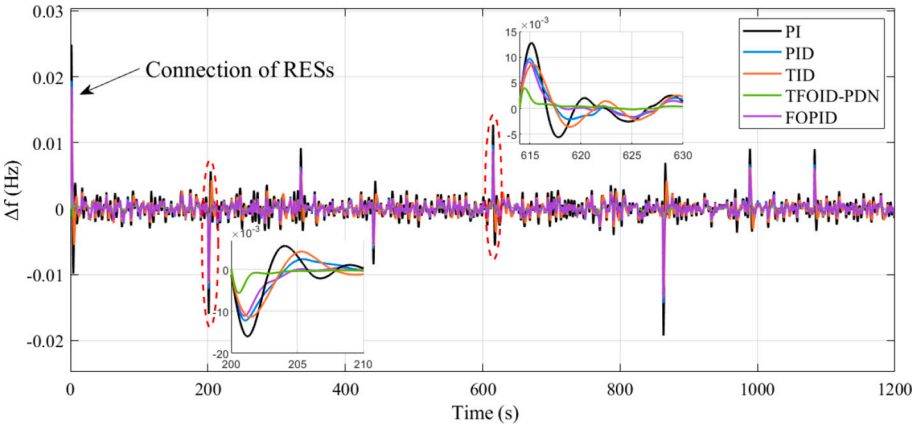


Fig. 22. System frequency deviation of EPS for scenario 2.

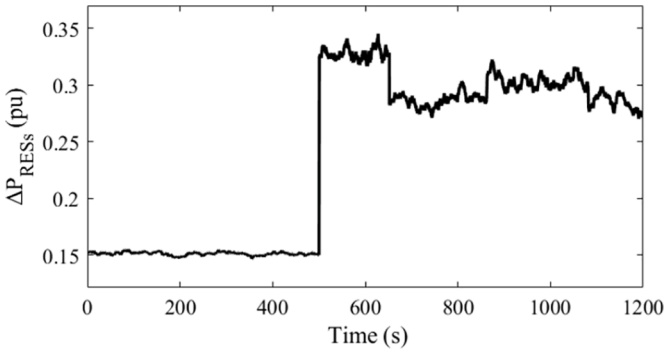


Fig. 23. RESs power variation for scenario 3.

function $G_{PI}(s)$ is presented in (33), and its block diagram representation is in Fig. 15. The EPS frequency deviation (i.e., Δf) is the input of the controller, and the output is the LFC control signal (i.e., ΔP_C) that is sent to dispatchable power plants. The limits of the parameters of PID and PI controllers are [0,10]

$$G_{PI}(s) = K_p + \frac{K_i}{s} \tag{33}$$

4.2. FOPID controller

Fractional-order (FO) controllers represent a modified version of the PID controller that is based on fractional calculus. The FOPID controller is a type of FO controllers where the order of integration and differentiation can be non-integer [52,53]. The transfer function of FOPID controller $G_{FOPID}(s)$ is provided in (34), and the corresponding block diagram is in Fig. 16. The EPS frequency deviation (i.e., Δf) is the input of the controller, and the output is the LFC control signal (i.e., ΔP_C) that is sent to dispatchable power plants. The limits of the gains (i.e., K_p, K_i, and K_d) are [0, 10], and the FO operators (i.e., λ and μ) are [0, 1].

$$G_{FOPID}(s) = K_p + \frac{K_i}{s^{\lambda}} + K_d s^{\mu} \tag{34}$$

4.3. TID controller

TID controller is a type of FO controllers [52,53] and is similar to the PID controller in the integral and derivative loops. The only difference is in the proportional loop, where the proportional gain K_p is replaced by the tilt term (K_t *$1/s^{(1/n)}$), where K_t is the tilt gain, and n is the tilt parameter. The TID controller transfer function $G_{TID}(s)$ is presented in (35), and its block diagram representation is in Fig. 17. The limits of the TID controller gains (i.e., K_t, K_i, and K_d) are [0, 10], and the tilt parameter n is [1,10]. The EPS frequency deviation (i.e., Δf) is the input of the controller, and the output is the LFC control signal (i.e., ΔP_C) that is sent to dispatchable power plants.

$$G_{TID}(s) = \frac{K_t}{s^{1/n}} + \frac{K_i}{s} + K_d s \tag{35}$$

4.4. Proposed two-stage TFOID-PDN controller

Fractional order controllers and PID controllers showed good performance in the LFC of different power systems. However, better performance could be obtained using two-stage controllers. Many studies

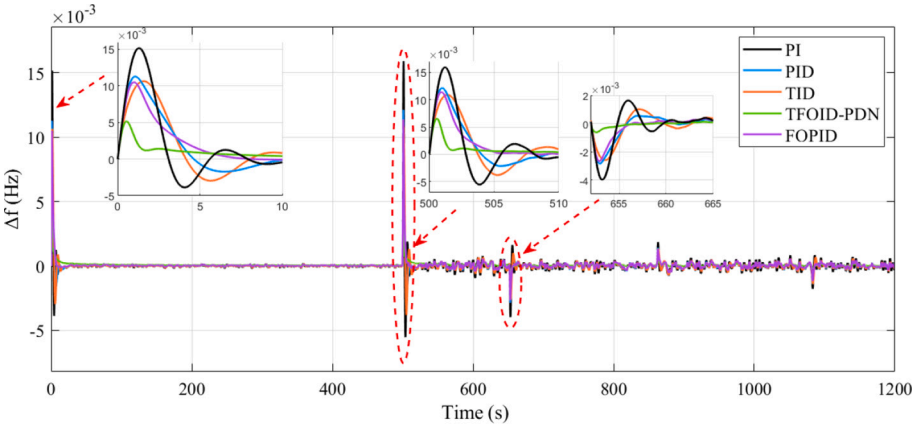


Fig. 24. System frequency deviation of EPS for scenario 3.

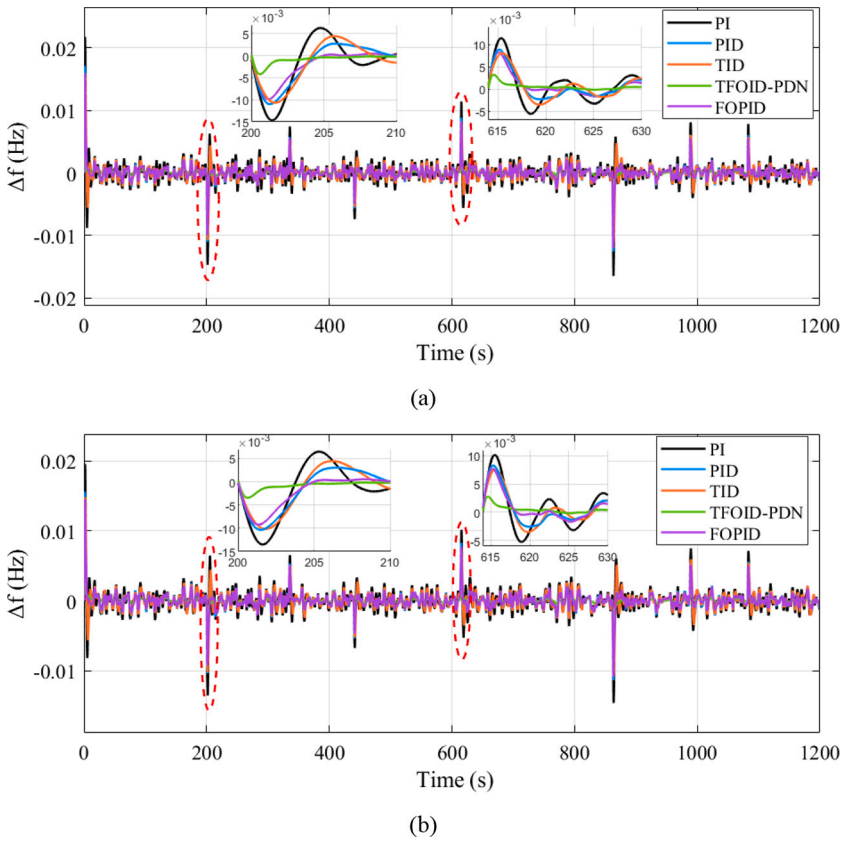


Fig. 25. System frequency deviation of EPS for scenario 4 with system inertia (H) variation; (a) +30 % of H, and (b) +60 % of H.

proposed two-stage controllers for LFC. For instance, Ref. [54] proposed a novel two-stage controller for frequency control of a microgrid with various RESs, controllable loads, and fuel cells. The proposed controller showed a better performance than PID and PID with a filter in the derivative branch. Another study [55] proposed a new two-stage controller for frequency control of two-area interconnected microgrids containing diverse RESs. The proposed controller showed superior performance over FOPID and PID controllers. This study proposes a novel two-stage controller called tilt FO integral derivative-proportional derivative with filter (TFOID-PDN). The first stage consists of the tilt branch of the TID controller and the integral and the derivative branches of the FOPID controller. The second stage contains the proportional and derivative branches of the PID controller. The proposed controller transfer function $G_{TFOID\text{-}PDN}(s)$ is presented in (36), and its block diagram representation is in Fig. 18. The limits of the controller parameters (i.e., K_t, K_i, K_{d1}, K_p, K_{d2}, and N) are [0, 10], the tilt parameter n is [1,10], and the fractional order operators (i.e., λ and μ) are [0, 1]. The EPS frequency deviation (i.e., Δf) is the input of the controller, and the output is the LFC control signal (i.e., ΔP_C) that is sent to dispatchable power plants.

$$G_{TFOID-PDN}(s) = \left(\frac{K_t}{s^{\frac{1}{n}}} + \frac{K_i}{s^{\lambda}} + K_{d1}s^{\mu}\right) * \left(K_p + K_{d2}\frac{N}{1+N\frac{1}{s}}\right) \tag{36}$$

4.5. Objective function of the Egyptian power system

The ISE objective function is used to tune the PID controller for the tested two-area power system. Similarly, in the EPS, the ISE objective function is used to tune the parameters of the proposed controller and the other controllers (i.e., FOPID, TID, PID, and PI). The ISE for the EPS is defined as given in (37). The proposed AHA optimization algorithm is used to find the optimal values of the controllers' parameters to achieve the optimal frequency control performance by minimizing the ISE objective function. The constraints of the optimization problem are the limits of the controllers' parameters. Equations of the controllers' limits are given in (38) to (46). Table 5 summarizes the optimization variables limits for the EPS. A 0.2pu SLP is applied at time $t = 20$s, and AHA is used to tune the parameters of the different studied controllers to minimize ISE in an iterative process until the maximum number of iterations is reached. The values of objective functions for different controllers and the corresponding optimal parameters of controllers are provided in Table 6. The convergence curves of AHA for different controllers are provided in Fig. 19. Table 6 and Fig. 19 show that the proposed TFOID-PDN achieved a much lower objective function value than the other controllers.

$$ISE = \int_0^{tsim} (\Delta f)^2 dt \tag{37}$$

$$K_p^{min} \le K_p \le K_p^{\max} \tag{38}$$

$$K_i^{min} \le K_i \le K_i^{\max} \tag{39}$$

$$K_{d1}^{min} \le K_{d1} \le K_{d1}^{\max} \tag{40}$$

$$K_{d2}^{min} \le K_{d2} \le K_{d2}^{\max} \tag{41}$$

$$K_t^{min} \le K_t \le K_t^{\max} \tag{42}$$

$$\lambda^{min} \le \lambda \le \lambda^{max} \tag{43}$$

$$\mu^{min} \le \mu \le \mu^{max} \tag{44}$$

$$n^{min} \le n \le n^{max} \tag{45}$$

$$N^{min} \le N \le N^{max} \tag{46}$$

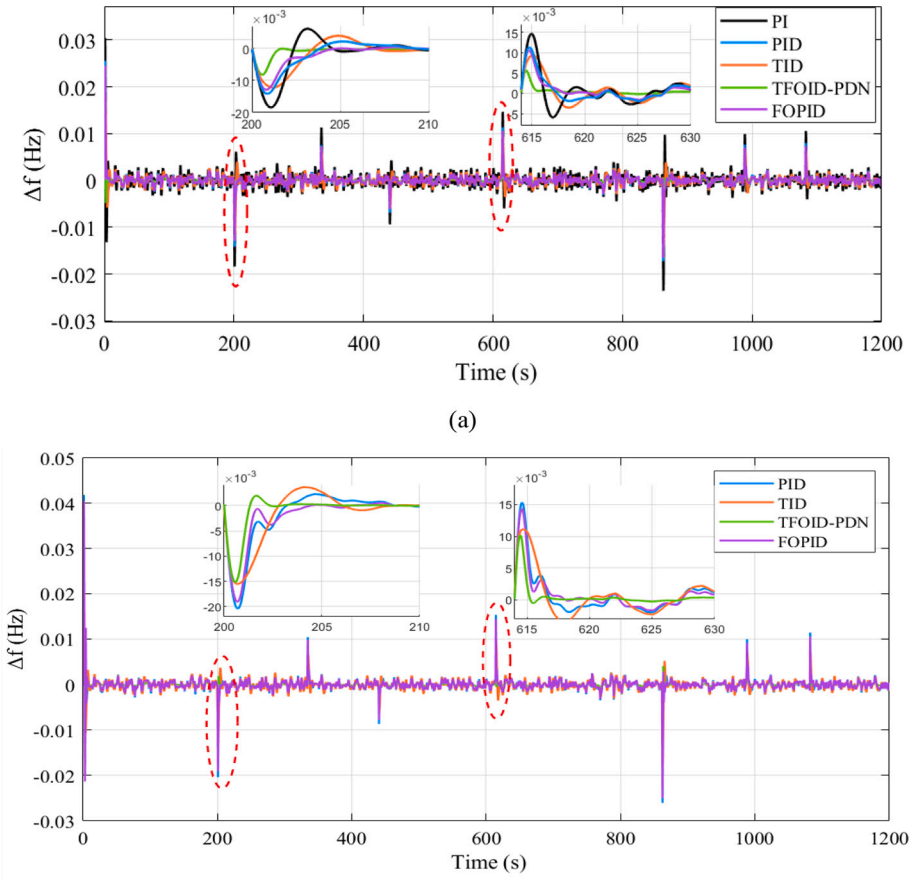


Fig. 26. System frequency deviation of EPS for scenario 4 with system inertia (H) variation; (a) -30 % of H, and (b) -60 % of H.

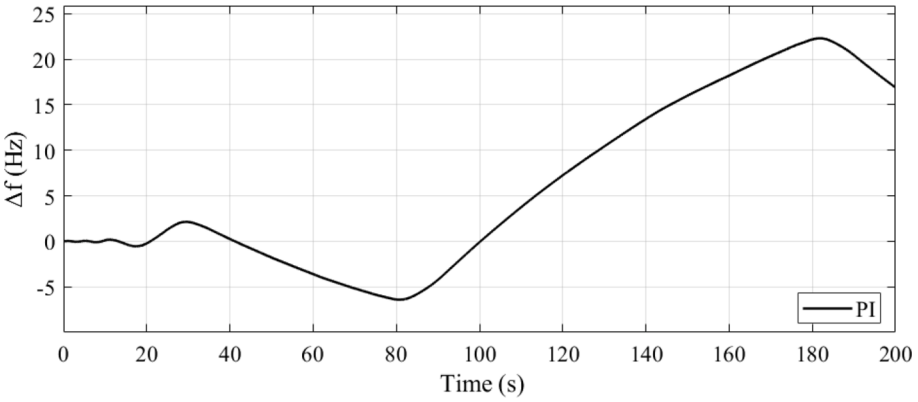


Fig. 27. System frequency deviation of EPS with PI controller for scenario 4 with −60 % change of H.

Table 7
Optimal parameters of the PI controller, PI+VI, and PI+VSG.

Controller	PI	PI + VI	PI + VSG
Parameters	$K_p = 9.9999$ $K_i = 9.8301$ – –	$K_p = 9.8659$ $K_i = 9.8044$ $K_{VI} = 4.9621$ –	$K_p = 9.9997$ $K_i = 9.9978$ $K_{VI} = 4.9615$ $K_D = 4.9439$
ISE	15.206e-4	9.749e-4	**7.577e-4**

5. Results and discussion

This section presents the results of the EPS subjected to different disturbances and operating conditions. The performance of different controllers is compared with the proposed two-stage controller. Additionally, the effect of VSG in enhancing the frequency stability of the EPS is studied. MATLAB/Simulink is used to model the EPS and conduct all simulations.

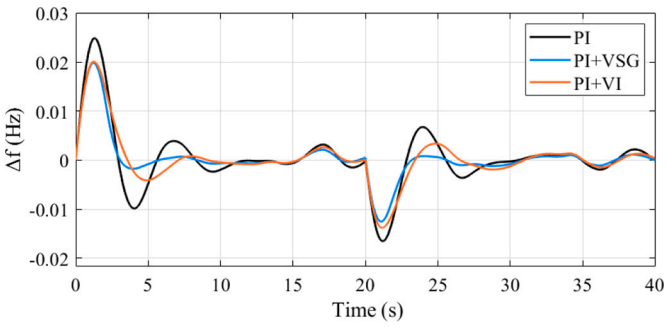


FIG. 28. System frequency deviation of EPS for case 1 of scenario 5.

5.1. Scenario 1: Step load change

In this scenario, a + 0.2 pu increase in load occurs in the EPS at time t = 20s and the value of frequency change is recorded. Fig. 20 shows the EPS frequency deviation for different controllers used for the LFC. There is an increase in the frequency at time $t = 0$ s due to the generation of intermittent RESs at the beginning of the simulation. The frequency deviation due to the connection of RESs happens in all the studied scenarios of EPS. The results show a superior performance of the proposed TFOID-PDN controller over FOPID, TID, PID, and PI controllers since it achieve the lowest frequency deviations and settling time. The other controllers show satisfactory performance, while the PI controller shows the worst performance with the highest frequency deviations and settling time.

5.2. Scenario 2: Random load change

In this scenario, a random step increases and decreases in load occur in the EPS, as shown in Fig. 21. The corresponding change in system frequency is given in Fig. 22 for all controllers. The proposed two-stage controller achieves the best performance for all changes in load with the lowest frequency deviations and settling times for all changes in load. The other controllers (i.e., FOPID, TID, and PID) show satisfactory performance, and the PI controller shows the worst performance with the highest frequency deviations and settling time.

5.3. Scenario 3: Large change in RESs generation

The RESs generation is intermittent and difficult to predict and could lead to a high mismatch between supply and demand and, consequently, large frequency deviations. Therefore, in this scenario, the effect of a large change in RESs generation on the EPS dynamic response is studied. Fig. 23 shows the changes in power generated by PV and WG (i.e., RESs). Fig. 24 shows the frequency change due to these disturbances for all controllers. The results show that the proposed two-stage controller can achieve small frequency deviations and obtain the best dynamic response compared to the other controllers.

5.4. Scenario 4: System uncertainty

With the increasing penetration of RESs and decreasing penetration of synchronous generators, the power system inertia constant (i.e., H) decreases, resulting in high-frequency deviations and high RoCoF after disturbances which could lead to power system instability. Additionally, the optimized frequency controller could provide unsatisfactory performance when the power system parameters change and should be retuned (i.e., find new optimal parameters of the controller) considering

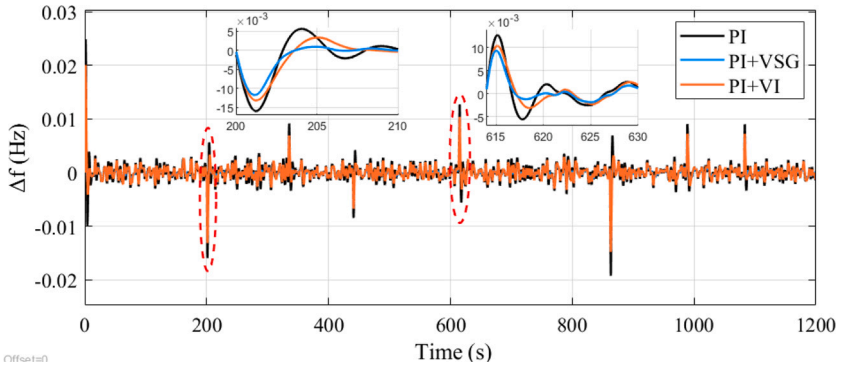


Fig. 29. System frequency deviation of EPS for case 2 of scenario 5.

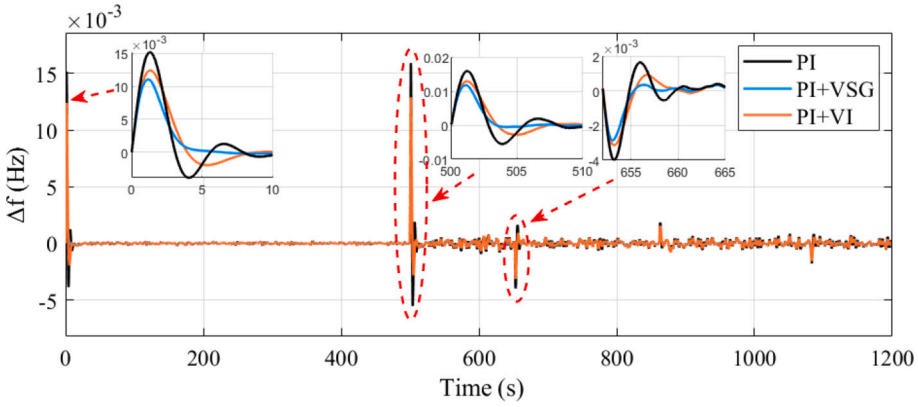


Fig. 30. System frequency deviation of EPS for case 3 of scenario 5.

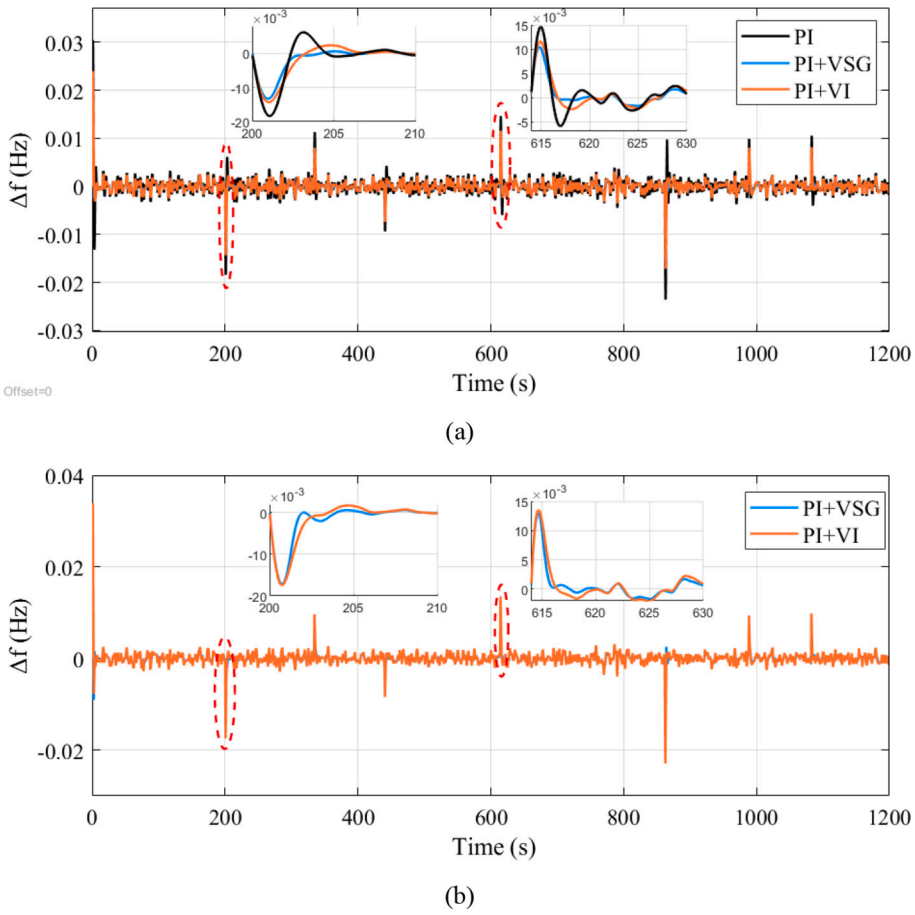


Fig. 31. System frequency deviation of EPS for scenario 5 of system inertia (H) variation; (a) -30 % of H, and (b) -60 % of H.

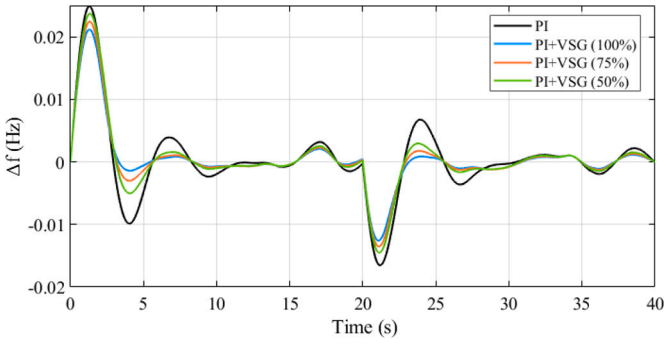


Fig. 32. System frequency deviation of EPS under case 5 of scenario 5 considering different aggregated EV capacities.

the new power system conditions. Therefore, in this scenario, the performance of all controllers is compared by considering changing of system inertia constant (H) by ±30 % and ± 60 % without re-tuning the controllers. A random step increase and decrease in load occur in the EPS, as shown in Fig. 21. The dynamic response presented in Fig. 25 shows that with the increase of H (i.e., +30 % and + 60 %) all controllers provide satisfactory performance, where the proposed controller provides the best response compared to the other controllers while the PI provides the poorest performance. The dynamic response presented in Fig. 26 shows that with the decrease in H (i.e., −30 % and − 60 %), the proposed controller provides the best response among other controllers, and the FOPID, TID, and PID controllers provide good performance. However, the PI controller could not keep the frequency within acceptable limits and resulted in frequency instability when H decreased by −60 % as shown in Fig. 27.

The satisfactory performance of the optimized controllers proves their robustness and insensitivity to changes in power system parameters and operating conditions. Moreover, it demonstrates the effectiveness of the proposed AHA optimization algorithm in finding optimal parameters of controllers that achieve a good performance in LFC applications.

5.5. Scenario 5: Enhancing EPS frequency stability using VSG

The previous subsection showed that the PI controller provided the worst performance by obtaining the highest frequency deviations, oscillations, and settling time, in the case of system uncertainty. Therefore, in this scenario, aggregated EVs are controlled to operate as VSG to enhance frequency stability by emulating synchronous generator characteristics (i.e., inertia (H) and damping (D) properties). Most of the studies in the literature consider constant values for the virtual inertia and damping gains [56]. In this study, optimal values of the VSG gains and PI controller gains are found using the proposed AHA optimization algorithm. The optimal values of gains are provided in Table 7. The VSG-based approach is compared with the base case where the PI controller is used for LFC and without VSG. Additionally, it is compared with the virtual inertia control (VIC) approach, where only the inertia emulation loop is considered.

To test the effect of VSG in enhancing the EPS frequency stability, many cases are studied considering different operating conditions and disturbances. In case 1, a + 0.2 pu step load change occurs in the EPS at time t = 20s. Fig. 28 shows that VSG can enhance the EPS frequency stability and obtain lower frequency deviations compared to the LFC

based on the PI controller with/without VIC. Note that the increase in the frequency at time $t = 0$ s is due to the generation of intermittent RESs at the beginning of the simulation.

Additionally, in case 2, various step increases and decrease of load occurs in the EPS, as shown in Fig. 21. The results provided in Fig. 29 prove the effectiveness of VSG in improving frequency stability, where it shows lower frequency deviations and settling times. In case 3, large changes in PV and WG power generation are considered, as shown in Fig. 23. The results presented in Fig. 30 show the ability of the proposed VSG to reduce frequency deviations when there is a large change in RESs generation.

In case 4, various step increases and decreases of load occur in the EPS, as shown in Fig. 21, and the value of H is decreased by −30 % and − 60 %. Fig. 31 shows that the presence of VSG could enhance the EPS frequency stability and obtain lower frequency deviations, oscillations, and settling time when the power system H decreased by −30 %. Fig. 27 in scenario 4 showed that the PI controller resulted in frequency instability when H decreases by −60 %. However, the use of VSG or VIC enhances the frequency stability of EPS and does not reach an instability state, as shown in Fig. 31(b).

Since EVs are not connected to the grid at all times because they are used for mobility needs, the EVs aggregated capacity changes. Case 5 studies the participation of VSG in frequency control considering the different aggregated capacities of EVs. 100 % refers to the situation of all the considered EVs are connected to the grid. Other percentages refer to a lower number of EVs connected to the grid. A + 0.2 pu step load change occurs in the EPS at time t = 20s. Fig. 32 shows that any percentage of aggregated EVs capacity controlled as VSG could improve the frequency stability compared to PI control without VSG. The increase of aggregated capacity of EVs improves the frequency stability.

6. Conclusion

Power systems are undergoing a fast transition from dispatchable conventional generation to intermittent renewable energy-based generation. This results in many challenges in frequency regulation. Therefore, recent studies proposed new controllers, optimization algorithms, and the participation of new flexible devices for enhancing the frequency stability of low inertia power systems. This article studied the performance of the recent artificial hummingbird algorithm (AHA) in load frequency control applications. The results prove that the proposed AHA provides better performance in terms of maximum undershoot, maximum overshoot, and settling time than the marine predators algorithm, grey wolf optimizer, and artificial bee colony optimization algorithms. Furthermore, the proposed new controller (TFOID-PDN) shows a superior performance over FOPID, TID, PID, and PI controllers considering different disturbances and operating conditions of the Egyptian power system (EPS). Moreover, the electric vehicles-based virtual synchronous generator (VSG) is effective in improving the frequency stability of EPS operating with a high share of renewable energy sources and low inertia. In the presence of VSG, lower frequency deviations, oscillations, and settling time are achieved. Furthermore, the VSG can prevent reaching the frequency instability state when the EPS is operating at low inertia.

CRediT authorship contribution statement

Morsy Nour: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing - original draft, Writing – review & editing. **Gaber Magdy:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Writing - original draft, Writing – review & editing. **José Pablo Chaves-Ávila:** Funding acquisition, Project administration, Resources, Supervision, Validation, Writing – review & editing. **Álvaro Sánchez-Miralles:** Funding acquisition, Project administration, Resources, Supervision, Writing – review & editing. **Francisco Jurado:** Data curation, Formal analysis, Funding acquisition, Investigation, Project administration, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Appendix 1

Parameters of two-area power system [38,39].

$K_{ps} = 120$ Hz/pu MW, $T_{ps} = 20$ s, $T_g = 0.08$ s, $T_t = 0.3$ s, $k_r = 0.5$, $T_r = 10$ s, $R = 2.4$ Hz/pu MW, $B = 0.425$ pu MW/Hz, and $T_{tie} = 0.086$.

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